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Topology optimization of hyperelastic structures with anisotropic fiber reinforcement under large deformations

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Abstract

Fiber-reinforced soft materials have emerged as promising candidates in various applications such as soft robotics and soft fibrous tissues. To enable a systematic approach to design fiber-reinforced materials and structures, we propose a general topology optimization framework for the computational optimized design of hyperelastic structures with nonlinear and anisotropic fiber reinforcements under large deformations. This framework simultaneously optimizes both the material distribution in the matrix phase and the orientations of the underlying fiber reinforcements, by parameterizing matrix and fiber phases individually using two sets of design variables. The optimized distribution of fiber orientations is chosen from a set of discrete orientations defined *a priori*, and several fiber orientation interpolation schemes are studied. In addition, this work proposes a novel anisotropic material interpolation scheme, which integrates both matrix and fiber design variables (both with material nonlinearity) into the stored-energy function. To improve the computational efficiency of both optimization and nonlinear structural analysis, we derive a fully decoupled fiber-matrix update scheme that performs parallel updates of the matrix and fiber design variables and employ the virtual element method (VEM) together with a tailored mesh adaptivity scheme to solve the finite elasticity boundary value problem. Design examples involving three objective functions are presented, demonstrating the efficiency and effectiveness of the proposed framework in designing anisotropic hyperelastic structures under large deformations.
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Keywords: Fiber-reinforced soft material; Topology optimization under large deformations; Decoupled design update scheme; Anisotropic hyperelasticity; Fiber-orientation interpolation; Virtual element method (VEM)

1. Introduction

Fiber-reinforced soft materials find potential applications in many emerging engineering fields, such as soft actuators [1–3], soft robotics [4,5] and soft tissue engineering [6,7]. As a powerful computational design tool,

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topology optimization offers a systematic approach to design fiber-reinforced soft materials to realize their multi-functionality. In general, topology optimization formulations involving fiber orientation design can be categorized into two types. In the first type, the fiber orientation at each point can continuously vary from $\theta_{\min}$ (e.g., 0) to $\theta_{\max}$ (e.g., π), see, e.g., [8–12]. The second type aims to find the best distribution of fiber angles among a discrete set of fiber orientations defined *a priori*. In this work, we focus on formulations of the second type. In order to ensure that the fiber orientation converges to a discrete set of candidate angles, a fiber orientation interpolation scheme is typically needed. Several fiber interpolation schemes have been proposed in the literature. The representative schemes include discrete material optimization (DMO) [13], shape function with penalization (SFP) [14] and bi-value coding parameterization (BCP) [15], and normal distribution function optimization (NDFO) and its variants [16,17]. Despite the above-mentioned progress, establishing a topology optimization formulation for designing fiber-reinforced soft materials and structures still needs further development. While the most studies have been limited to linear elastic behavior (for both matrix and fiber), formulations capable of accounting for nonlinear behaviors of both matrix and fiber phases under large deformations and simultaneously optimizing both phases are rarely considered (see, however, Refs. [18–20] for work on simultaneous optimization of linear matrix and fiber phases, and Ref. [21] for work on optimizing fiber orientations in (non-designable) hyperelastic matrix under large deformations).

From the optimization perspective, topology optimization considering multiple pre-defined candidate fiber orientations naturally involves multiple sets of design variables, making the design update process a computationally expensive task for general-purpose update schemes. In Refs. [22,23], a design variable update scheme, named the Zhang–Paulino–Ramos (ZPR), was proposed considering multiple materials with both linear and nonlinear properties. The ZPR exploits the separable feature of the convex approximated subproblem at each optimization step and derives a decoupled update formula for the design variables associated with each volume constraint. The ZPR update has been mostly used in multi-material topology optimization consisting of multiple volume constraints and has not been explored in topology optimization problems with matrix phase and fiber reinforcements.

In this work, we put forth a general topology optimization framework to design hyperelastic structures with anisotropic fiber reinforcements under large deformations. The contributions of our proposed framework are as follows:

- First, the propsoed framework simultaneously optimizes both material distribution in matrix phase and orientation distribution of the underlying fiber reinforcements under large deformations. In addition, we take into account nonlinear behaviors of both matrix and fiber phases.
- Second, we introduce a design parameterization, where the matrix and fiber phases are represented by separate design variables. The advantage of this parameterization is that it allows for the seamless incorporation of geometric restrictions (e.g., symmetry constraints) into the matrix phase without influencing the fiber orientation distributions. To integrate the representations of matrix and fiber phases, we propose a general anisotropic material interpolation scheme that relates the stored-energy function of the resulting composites to the design variables associated with both phases.
- Third, inspired by the ZPR design update scheme for multi-material topology optimization [22], a decoupled design variable update scheme is proposed to enable separable and parallelizable update of the design variables associated with the density field and each fiber orientation field. We further derive explicit update formula (i.e., no inner iterations are needed) for the design variables associated with each fiber orientation field.
- Last but not the least, we adopt the virtual element method (VEM) in conjunction with a tailored adaptive refinement and coarsening scheme [24,25] to efficiently solve the boundary value problem of finite elasticity. As an emerging computational method, the VEM has attracted growing interests in the fields of computational mechanics, see, e.g., [26–32], and topology optimization [33–35]. The VEM has a unique feature of handing arbitrary element geometries and, thus, offers an ideal platform for flexible and adaptive refinement and coarsening schemes to accelerate topology optimization without sacrificing design resolution.

The remainder of this paper is organized as follows. Section 2 briefly reviews the constitutive models for transversely isotropic materials and the variational principle of finite elasticity. Section 3 introduces the VEM formulation to simulate the nonlinear elastic response of fiber-reinforced material under large deformations, including a simple example to verify the convergence of the VEM approximation. In Section 4, we introduce the general topology optimization framework for the design of fiber-reinforced hyperelastic structures with emphasis

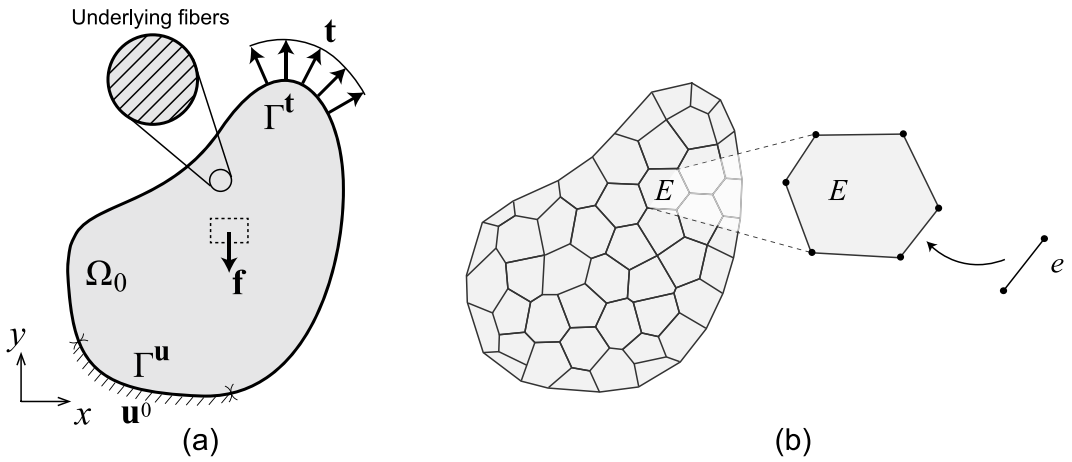


Fig. 1. (a) A schematic illustration of domain and boundary conditions for the solid Ω_0. (b) Schematic illustrations of the discretized domain, a generic element E, and a generic edge e of the element E.

on design parameterization, the fiber orientation and anisotropic material interpolation schemes, and the energy interpolation approach. In Section 5, a decoupled design variable update scheme is proposed and explicit update formulas are derived for the design variables associated with the fiber orientation fields. Section 6 presents design problems in conjunction with three objective functions and two fiber interpolation schemes to demonstrate the effectiveness of the proposed topology optimization framework. Section 7 provides several concluding remarks of the present work. The sensitivity analysis of the proposed topology optimization framework is provided in Appendix.

2. Finite elasticity considering transversely isotropic solids

This section briefly reviews the constitutive models of transversely isotropic materials [36] and the variational formulation adopted in this work. We adopt a Lagrangian description and neglect the presence of body force. As illustrated in Fig. 1(a), let us consider a solid that occupies Ω_0 in its undeformed configuration (which is assumed to be stress-free). On the boundary, a traction field $\mathbf{t}$ and a displacement field $\mathbf{u}^0$ are applied on $\Gamma^{\mathbf{t}}$ and $\Gamma^{\mathbf{u}}$, respectively, such that $\Gamma^{\mathbf{t}} \cup \Gamma^{\mathbf{u}} = \partial\Omega_0$ and $\Gamma^{\mathbf{t}} \cap \Gamma^{\mathbf{u}} = \emptyset$. Under the applied traction and displacement fields, the solid experiences a deformation map $\boldsymbol{\chi} : \Omega_0 \to \mathbb{R}^d$ (d is the dimension) which transforms a material point $\mathbf{X} \in \Omega_0$ to a position $\boldsymbol{x} = \boldsymbol{\chi}(\mathbf{X}) \in \Omega$ in the deformed configuration Ω. The deformation gradient field $\mathbf{F}$ is given by $\mathbf{F} = \partial\boldsymbol{\chi}(\mathbf{X})/\partial\mathbf{X}$.

We assume that the solid Ω_0 is transversely isotropic and consists of a compressible and isotropic matrix phase reinforced by fibers whose orientation is given by unit vector $\mathbf{a}_0$ in the undeformed configuration. The nonlinear elastic behavior of this hyperelastic solid is characterized by a stored-energy function Ψ, which depends on the five invariants of the right Cauchy-Green deformtion tensor $\mathbf{C} = \mathbf{F}^T\mathbf{F}$ [37,38]:

$$I_1 = \mathrm{tr}(\mathbf{C}), \quad I_2 = \frac{1}{2}\Big(\mathrm{tr}(\mathbf{C})^2 - \mathrm{tr}(\mathbf{C}^2)\Big), \quad J = \sqrt{\det\mathbf{C}}, \quad I_4 = \mathbf{a}_0 \cdot \mathbf{C}\mathbf{a}_0, \quad I_5 = \mathbf{a}_0 \cdot \mathbf{C}^2\mathbf{a}_0. \tag{1}$$

Furthermore, neglecting the dependence of the stored-energy function Ψ on I_5 [37,38], Ψ takes the form of

$$\Psi(I_1, I_2, J, I_4) = \Psi_{\mathrm{M}}(I_1, I_2, J) + \Psi_{\mathrm{F}}(I_4), \tag{2}$$

where Ψ_{M} and Ψ_{F} are the contributions from the matrix and fiber phases, respectively. In this work, we assume the matrix phase is made up of a compressible Neo-Hookean material characterized by

$$\Psi_{\mathrm{M}}(I_1, I_2, J) = \frac{\mu}{2}(I_1 - 3) - \mu \ln J + \frac{\kappa}{2}(J-1)^2, \tag{3}$$

where μ and κ are the initial shear and bulk moduli, respectively. Other isotropic hyperelastic models, such as the Ogden model [39] and the Lopez–Pamies model [40], can be adopted as well. For the contribution from the fibers, we use the following model [41]

$$\Psi_{\mathrm{F}}(I_4) = \begin{cases} \frac{E_t}{2}(\sqrt{I_4} - 1)^2 & \text{if } I_4 \geq 1, \\ \frac{E_c}{2}(\sqrt{I_4} - 1)^2 & \text{if } I_4 < 1, \end{cases} \tag{4}$$

where E_t and E_c are the moduli of fibers in tension and compression, respectively. We note that we use different moduli for tension and compression to account for the material nonlinearity of fibers as they typically behave differently in tension and compression. Other forms of Ψ_{F} such as the exponential one proposed in [37] can also be adopted.

We use the second Piola–Kirchoff (PK) stress as the stress measure. According to the stored-energy function, the second PK stress $\mathbf{S}$ is given by

$$\mathbf{S} = 2\frac{\partial \Psi}{\partial \mathbf{C}}(I_1, I_2, J, I_4) = 2\frac{\partial \Psi_{\mathrm{M}}}{\partial \mathbf{C}}(I_1, I_2, J) + 2\frac{\partial \Psi_{\mathrm{F}}}{\partial \mathbf{C}}(I_4) = \mathbf{S}_{\mathrm{M}} + \mathbf{S}_{\mathrm{F}}, \tag{5}$$

where $\mathbf{S}_{\mathrm{M}}$ and $\mathbf{S}_{\mathrm{F}}$ are the contributions from the matrix and fiber phases, respectively. Applying chain rule to the above expression and plugging in the expressions of the invariants in (1) yield

$$\mathbf{S} = 2\left[\frac{\partial \Psi_{\mathrm{M}}}{\partial I_1}\mathbf{I} + \frac{\partial \Psi_{\mathrm{M}}}{\partial I_2}\big(I_1\mathbf{I} - \mathbf{C}\big) + \frac{\partial \Psi_{\mathrm{M}}}{2\partial J}J\mathbf{C}^{-1} + \frac{\partial \Psi_{\mathrm{F}}}{\partial I_4}\mathbf{a}_0 \otimes \mathbf{a}_0\right]. \tag{6}$$

We adopt a displacement-based variational principle to describe the boundary value problem of finite elasticity. The displacement field is the only independent field. The deformation gradient $\mathbf{F}$, the right Cauchy–Green deformation tensor $\mathbf{C}$, and the second PK stress tensor $\mathbf{S}$ are dependent on the displacement field. According to the Principle of Minimum Potential Energy, the displacement $\mathbf{u}$ in the equilibrium state is the one that minimizes the potential energy $\Pi(\cdot)$ among all kinematically admissible displacements, namely,

$$\mathbf{u} = \arg\min_{\mathbf{v}\in\mathcal{K}} \Pi(\mathbf{v}), \tag{7}$$

where $\mathcal{K}$ stands for the space of kinematically admissible displacements, and the potential energy $\Pi(\mathbf{v})$ is given by

$$\Pi(\mathbf{v}) = \int_{\Omega_0}\Big[\Psi_{\mathrm{M}}\big(I_1(\mathbf{v}), I_2(\mathbf{v}), J(\mathbf{v})\big) + \Psi_{\mathrm{F}}\big(I_4(\mathbf{a}_0, \mathbf{v})\big)\Big]\mathrm{d}\mathbf{X} - \int_{\Gamma^{\mathbf{t}}}\mathbf{t}\cdot\mathbf{v}\mathrm{d}S. \tag{8}$$

The weak form of the Euler–Lagrange equation of (7) is given by

$$G(\mathbf{v}, \delta\mathbf{v}) = \int_{\Omega_0}\Big\{\mathbf{F}(\mathbf{v})\Big[\mathbf{S}_{\mathrm{M}}(\mathbf{v}) + \mathbf{S}_{\mathrm{F}}(\mathbf{a}_0, \mathbf{v})\Big]\Big\} : \nabla(\delta\mathbf{v})\mathrm{d}\mathbf{X} - \int_{\Gamma^{\mathbf{t}}}\mathbf{t}\cdot\delta\mathbf{v}\mathrm{d}S = 0 \quad \forall\delta\mathbf{v}\in\mathcal{K}_0, \tag{9}$$

where $\delta\mathbf{v}$ is the variation of $\mathbf{v}$, and $\mathcal{K}_0$ is the set of adimissible displacements whose values vanish on $\Gamma^{\mathbf{u}}$.

3. The VEM formulation for modeling fiber-reinforced elastomers

Consider a discretization of the domain Ω_0 into non-overlapping elements. By convention, we denote the discretized domain as Ω_h. We also use e and E to denote a generic edge and element, respectively, with $|e|$ and $|E|$ denoting the length and area of e and E, respectively. An illustration of the discretized domain Ω_h, a generic element E and a generic edge e is provided in Fig. 1(b). We adopt the VEM to discretize and approximate the weak form (9) [42,43]. Alternatively, one can also use the finite element method (FEM) [44,45]. Hereafter, we address the local VEM space [42], the VEM projection, and the construction of a first-order VEM approximation to the continuous variational principle. A study is presented at the end of this section to demonstrate the convergence of the VEM approximation.

3.1. Local VEM space and projection

Consider a generic polygon E with n vertices, located at $\mathbf{X}_i = [X_i, Y_i]^T$, $i = 1, \ldots, n$, in the undeformed configuration numbered in a counterclockwise fashion. The local displacement space $\boldsymbol{\mathcal{V}}(E)$ is defined as:

$$\boldsymbol{\mathcal{V}}(E) = \Big\{\mathbf{v}\in[\mathcal{H}_1(E)]^2 : \mathbf{v}|_e \in [\mathcal{P}_1(e)]^2\ \forall e\in\partial E \text{ and } \boldsymbol{\Delta}\mathbf{v} = \mathbf{0} \text{ in } E\Big\}, \tag{10}$$

where $\mathcal{P}_1(\cdot)$ is the space of linear functions and $\boldsymbol{\Delta}$ stands for the vector Laplacian operator. According to the above definition, the local displacement space $\boldsymbol{\mathcal{V}}(E)$ is linearly complete, namely $[\mathcal{P}_1(E)]^2 \subseteq \boldsymbol{\mathcal{V}}(E)$, and thus has good approximation properties. We can show that the dimension of $\boldsymbol{\mathcal{V}}(E)$ is $2n$, and the displacements at the vertices of E form a complete set of degrees of freedom (DOFs) of $\boldsymbol{\mathcal{V}}(E)$ [42].

A main feature of the local displacement space $\boldsymbol{\mathcal{V}}(E)$ is that it contains functions which are implicitly defined in the interior of E. Thus, instead of numerically integrating the weak form using the displacement field and its gradients at the integration points, the VEM introduces a projection operator $\boldsymbol{\Pi}_E^{\nabla} : \boldsymbol{\mathcal{V}}(E) \rightarrow [\mathcal{P}_1(E)]^2$, which maps the implicit displacement fields in E onto linear functions. For any given displacement $\mathbf{v} \in \boldsymbol{\mathcal{V}}(E)$, the projection $\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}$ is uniquely defined by the following two conditions:

$$\begin{cases} \int_E \nabla(\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}) : \nabla\mathbf{p}\mathrm{d}\mathbf{X} = \int_E \nabla\mathbf{v} : \nabla\mathbf{p}\mathrm{d}\mathbf{X} \quad \forall\mathbf{p} \in [\mathcal{P}_1(E)]^2, \quad \text{and} \\ \sum_{i=1}^m \big(\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}\big)(\mathbf{X}_i) = \sum_{i=1}^m \mathbf{v}(\mathbf{X}_i). \end{cases} \tag{11}$$

The first condition in the above definition determines the linear term of $\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}$. The second condition determines the constant term by enforcing that the summation of the values of $\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}$ evaluated at all the vertices of E equals to the summation of all the DOFs of $\mathbf{v}$. We note that the right-hand-sides of both conditions in (11) are exactly computable without knowing $\mathbf{v}$ in the interior of E. In particular, applying integration by parts to the right-hand-side on the first condition and realizing that $\nabla\mathbf{p}$ is a constant tensor, we have

$$\int_E \nabla\mathbf{v} : \nabla\mathbf{p}\mathrm{d}\mathbf{X} = \Big(\int_E \nabla\mathbf{v}\mathrm{d}\mathbf{X}\Big) : \nabla\mathbf{p} = \Big(\int_{\partial E} \mathbf{v}\otimes\mathbf{N}\mathrm{d}S\Big) : \nabla\mathbf{p} \quad \forall\mathbf{p} \in [\mathcal{P}_1(E)]^2, \tag{12}$$

where $\mathbf{N}$ is the outward normal vector of E. Because any function $\mathbf{v}$ in the local space $\boldsymbol{\mathcal{V}}(E)$ is piecewise linear and explicitly known on the boundary of E, the above equation is exactly computable when we know the DOFs of $\mathbf{v}$ and the geometric information of E.

3.2. A conforming displacement-based VEM approximation

The global VEM space $\mathcal{K}_h$ for the displacement field $\mathbf{v}_h$ is a finite dimensional subspace of $\mathcal{K}$ given by

$$\mathcal{K}_h = \Big\{\mathbf{v}_h \in \mathcal{K} : \mathbf{v}_h|_E \in \boldsymbol{\mathcal{V}}(E) \quad \forall E \in \Omega_h\Big\}. \tag{13}$$

Accordingly, a VEM approximation of the potential function $\Pi(\mathbf{v})$, which is denoted by $\Pi_h^{VEM}(\mathbf{v}_h)$, is constructed as

$$\begin{aligned} \Pi_h^{VEM}(\mathbf{v}_h) = \sum_{E\in\Omega_h} \Bigg\{ |E| \Big[\Psi_{\mathrm{M}}\Big(I_1(\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h), I_2(\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h), J(\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h)\Big) + \Psi_{\mathrm{F}}\Big(I_4(\mathbf{a}_0, \boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h)\Big)\Big] \\ + \frac{1}{2}\alpha_E(\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h) S_h^E(\mathbf{v}_h - \boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h, \mathbf{v}_h - \boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h) \Bigg\} - \langle \mathbf{t}, \mathbf{v}_h\rangle_h. \end{aligned} \tag{14}$$

Descriptions of each term on the right-hand-side of the above approximation are provided in sequence below. The first two terms are the consistency terms for the stored-energy contributions from matrix and fiber phases, respectively. For each element E in the mesh, both consistency terms are computed using the projected local displacement field $\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h$ defined in the previous subsection. The third term is a stabilization term which assigns energy to the non-physical spurious deformation modes in the consistency terms. We only stabilize the stored-energy function for the matrix phase, which has been shown to be sufficient in [43,46] for simulating fiber-reinforced hyperelastic solids. We adopt the trace-based stabilization proposed in [28], where the stabilization parameter $\alpha_E(\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h)$ is a scalar-valued function taken to be the trace of the tangent modulus tensor of Ψ_{M}. Since the matrix phase is assumed to be isotropic, an explicit expression of $\alpha_E(\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h)$ is given by [28]

$$\begin{aligned} \alpha_E(\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h) = \frac{1}{4}\Big[4I_1\frac{\partial^2\Psi_{\mathrm{M}}}{\partial I_1\partial I_1} + (8I_1I_2 - 4J^2I_1)\frac{\partial^2\Psi_{\mathrm{M}}}{\partial^2 I_2\partial I_2} + I_1\frac{\partial^2\Psi_{\mathrm{M}}}{\partial J\partial J} + 16I_2\frac{\partial^2\Psi_{\mathrm{M}}}{\partial I_1\partial I2} + 8J\frac{\partial^2\Psi_{\mathrm{M}}}{\partial I_1\partial J} \\ + 2JI_1\frac{\partial^2\Psi_{\mathrm{M}}}{\partial I_2\partial J} + 8\frac{\partial\Psi_{\mathrm{M}}}{\partial I_1} + 2I_1\frac{\partial\Psi_{\mathrm{M}}}{\partial I_2}\Big], \end{aligned} \tag{15}$$

where the dependence of I_1, I_2, and J on $\boldsymbol{\Pi}_E^{\nabla}\mathbf{v}_h$ are assumed in the above expression. The variable $S_h^E(\cdot,\cdot)$ is a bilinear form given by [42]

$$S_h^E(\mathbf{v},\mathbf{w}) = \sum_{v\in E} \mathbf{v}(\mathbf{X}_v)\cdot\mathbf{w}(\mathbf{X}_v) \quad \forall\mathbf{v},\mathbf{w} \in \boldsymbol{\mathcal{V}}(E), \tag{16}$$

where v stands for a generic vertex of E with $\mathbf{X}_v$ being its position vector in the undeformed configuration. The last term in (14) is the VEM approximation of the loading term which, based on [28], is given by

$$\langle \mathbf{t}, \mathbf{v}_h \rangle_h = \sum_{e \in \Gamma_h^{\mathbf{t}}} \sum_{v \in e} w_v^e \mathbf{t}(\mathbf{X}_v) \cdot \mathbf{v}_h(\mathbf{X}_v), \tag{17}$$

where $\Gamma_h^{\mathbf{t}}$ stands for the boundary of the mesh that the traction is applied and w_v^e is the weight of a first-order Gauss–Lobatto rule associated with the vertex $\mathbf{X}_v$ on edge e.

The discrete problem consists of seeking the equilibrium displacement $\mathbf{u}_h \in \mathcal{K}_h$ such that it minimizes Π_h^{VEM}, namely,

$$\mathbf{u}_h = \arg \min_{\mathbf{v}_h \in \mathcal{K}_h} \Pi_h^{VEM}(\mathbf{v}_h). \tag{18}$$

Accordingly, the VEM approximation of the weak form is given by

$$\begin{aligned} G_h(\mathbf{v}_h, \delta\mathbf{v}_h) = \sum_E \Big\{ & \Big[|E| (\mathbf{I} + \nabla \boldsymbol{\Pi}_E^{\nabla} \mathbf{v}_h) \Big(\mathbf{S}_{\mathrm{M}}(\boldsymbol{\Pi}_E^{\nabla} \mathbf{v}_h) + \mathbf{S}_{\mathrm{F}}(\mathbf{a}_0, \boldsymbol{\Pi}_E^{\nabla} \mathbf{v}_h) \Big) \\ & + \frac{1}{2} S_h^E(\mathbf{v}_h - \boldsymbol{\Pi}_E^{\nabla} \mathbf{v}_h, \mathbf{v}_h - \boldsymbol{\Pi}_E^{\nabla} \mathbf{v}_h) \frac{\partial \alpha_E}{\partial \mathbf{F}} (\boldsymbol{\Pi}_E^{\nabla} \mathbf{v}_h) \Big] : \nabla \boldsymbol{\Pi}_E^{\nabla} (\delta \mathbf{v}_h) \\ & + \alpha_E (\boldsymbol{\Pi}_E^{\nabla} \mathbf{v}_h) S_h^E(\mathbf{v}_h - \boldsymbol{\Pi}_E^{\nabla} \mathbf{v}_h, \delta \mathbf{v}_h - \boldsymbol{\Pi}_E^{\nabla} (\delta \mathbf{v}_h)) \Big\} \\ & - \langle \mathbf{t}, \delta \mathbf{v}_h \rangle_h = 0 \quad \forall \delta \mathbf{v}_h \in \mathcal{K}_h^0. \end{aligned} \tag{19}$$

In our implementation, the Newton-type algorithms are used to solve the nonlinear system of equations using the second-order information of Π_h^{VEM}, whose expressions can be obtained by linearizing (19) using the standard procedure [47].

3.3. A convergence study

To verify the convergence and accuracy of the VEM approximation in simulating the nonlinear response of fiber-reinforced elastomers, we present a numerical study to demonstrate the convergence of the displacement-based VEM approximation under mesh refinement. We consider a boundary value problem shown in Fig. 2(a), where a rectangular block of size $1 \times \pi$ is bent into a semi-circle. The exact displacement solution $\mathbf{u} = [u_x, u_y]^T$ describing such a deformation is given by

$$u_x(X, Y) = -1 + (1 + X)\cos(Y) - X \quad u_y(X, Y) = (1 + X)\sin(Y) - Y. \tag{20}$$

In this study, we apply the exact displacement field on the entire boundary of the rectangular block together with a body force computed from the exact displacement solution and then evaluate the L^2 norm and H^1 semi-norm of the displacement errors, denoted by $\boldsymbol{\epsilon}_{\mathbf{u},0}$ and $\boldsymbol{\epsilon}_{\mathbf{u},1}$, respectively, via [28]

$$\begin{aligned} \boldsymbol{\epsilon}_{\mathbf{u},0} &= \sqrt{\sum_{E \in \Omega_h} \int_E \Big(\mathbf{u} - \boldsymbol{\Pi}_E^{\nabla} \mathbf{u}_h \Big) \cdot \Big(\mathbf{u} - \boldsymbol{\Pi}_E^{\nabla} \mathbf{u}_h \Big) \mathrm{d}\mathbf{X}} \quad \text{and} \\ \boldsymbol{\epsilon}_{\mathbf{u},1} &= \sqrt{\sum_{E \in \Omega_h} \int_E \Big(\nabla \mathbf{u} - \nabla(\boldsymbol{\Pi}_E^{\nabla} \mathbf{u}_h) \Big) : \Big(\nabla \mathbf{u} - \nabla(\boldsymbol{\Pi}_E^{\nabla} \mathbf{u}_h) \Big) \mathrm{d}\mathbf{X}}, \end{aligned} \tag{21}$$

where $\mathbf{u}_h$ is the VEM displacement solution and the integral within each element is evaluated using a 5th-order triangulation scheme.

We assume the matrix phase of the block to be characterized by the Neo-Hookean model described by the stored-energy function (3) with $\mu = 1,000$ and $\kappa = 3,000$. Four test cases are considered for the layout of the fiber reinforcements, which are shown in Fig. 2(b). In the first and second test cases, the matrix is reinforced by one family of fibers with orientation $\theta = 0$ and $\theta = 3\pi/4$, respectively, and the moduli of the fiber in tension and compression are set to be $E_t = E_c = 3,000$. In the third and fourth test cases, the matrix is reinforced by two families of fibers with orientation pairs $(\theta_1, \theta_2) = (0, \pi/2)$ and $(\theta_1, \theta_2) = (\pi/4, 3\pi/4)$, respectively. The tension

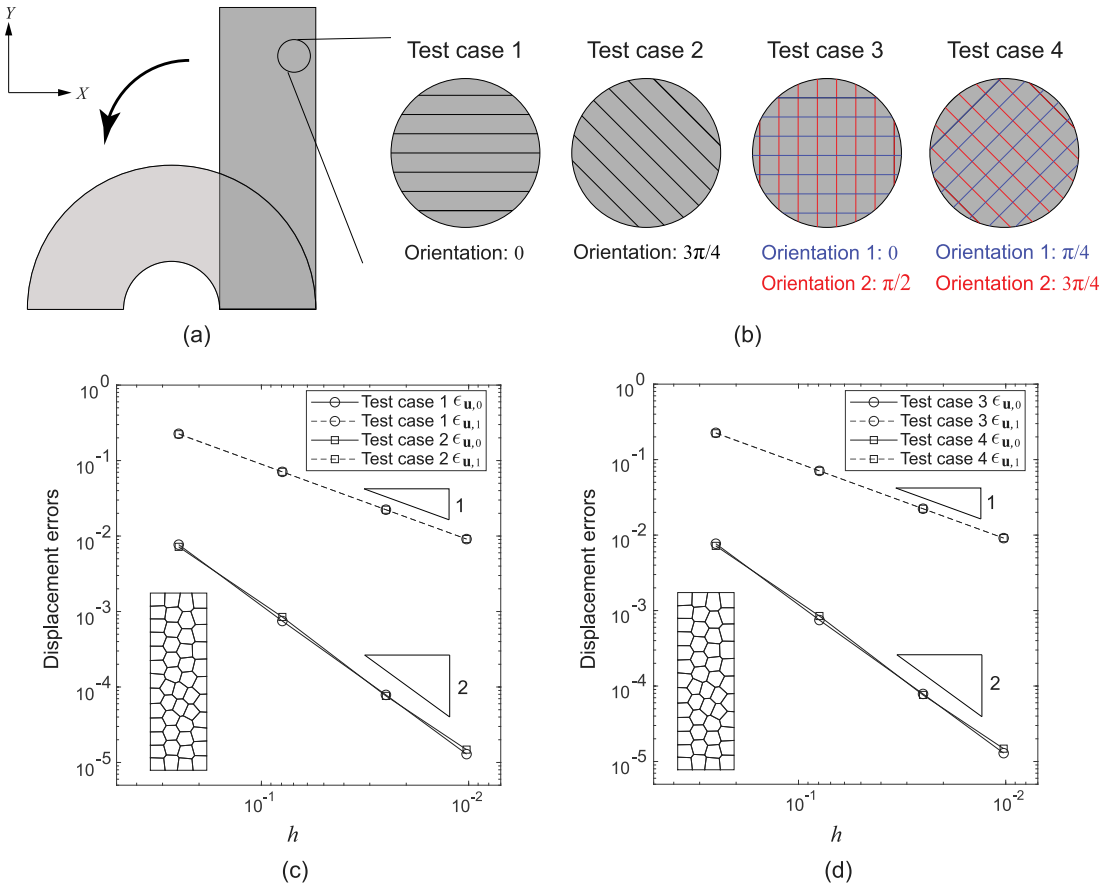


Fig. 2. (a) An illustration of the problem studied in the convergence test. (b) Schematic illustrations of the fiber reinforcement orientations in the four test cases. (c) Convergence of the displacement errors under mesh refinement in Test cases 1 and 2 with one fiber family. (d) Convergence of the displacement errors under mesh refinement in Test cases 3 and 4 with two fiber families.

and compression moduli of the fibers associated with orientations θ_1 and θ_2 are taken to be $E_t = E_c = 3,000$ and $E_t = E_c = 1,500$, respectively.

The Centroidal Voronoi Tessellation (CVT) meshes [48] are considered in this convergence study. An example of this mesh is shown in Fig. 2(c). In Fig. 2(c) and (d), we plot the convergence of errors of the VEM displacement solutions $\boldsymbol{\epsilon}_{\mathbf{u},0}$ and $\boldsymbol{\epsilon}_{\mathbf{u},1}$ as functions of the average element size h under mesh refinement for all the test cases. Each data point in the figure is computed by averaging the results from three trials. It is observed that optimal convergence rates are achieved for both the L^2 norm and H^1 semi-norm of the displacement errors, verifying that the VEM approximation is accurate and convergent in simulating the nonlinear elastic responses of fiber-reinforced solids under large deformations.

4. Topology optimization formulation for anisotropic fiber-reinforced hyperelastic structures under large deformations

We propose a general topology optimization framework for designing fiber-reinforced hyperelastic structures together with the associated design parameterization, matrix and fiber interpolation schemes, and the energy interpolation. The sensitivity analysis of the proposed topology optimization framework is presented in the Appendix.

4.1. A general topology optimization framework and design parameterization

We assume that the mesh Ω_h is composed of M elements and N nodes, and there are a total of m pre-defined candidate fiber orientations. The goal of the proposed optimization framework is to simultaneously find both the optimal distributions of matrix material and fiber orientation (among the m candidate orientations) that minimize a given objective function. Mathematically, we propose the following topology optimization formulation:

$$\begin{aligned}
&\min_{\boldsymbol{\rho},\boldsymbol{\xi}_k\in\mathbb{R}^M} J\Big(\boldsymbol{\rho},\boldsymbol{\xi}_1,\ldots,\boldsymbol{\xi}_{m_\xi},\mathbf{u}_h(\boldsymbol{\rho},\boldsymbol{\xi}_1,\ldots,\boldsymbol{\xi}_{m_\xi})\Big)\\
&\text{s.t.}\quad g_V(\boldsymbol{\rho})=\frac{\mathbf{V}^T\tilde{\boldsymbol{\rho}}}{|\Omega_h|}-V_{\max}\le 0\\
&\qquad \rho_{\min}\le\rho^{(e)}\le 1,\quad e=1,\ldots,M\\
&\qquad \xi_{\min}\le\xi_k^{(e)}\le\xi_{\max},\quad e=1,\ldots,M \text{ and } k=1,\ldots,m_\xi\\
&\text{with}\quad \mathbf{u}_h(\boldsymbol{\rho},\boldsymbol{\xi}_1,\ldots,\boldsymbol{\xi}_{m_\xi})=\arg\min_{\mathbf{v}_h\in\mathcal{K}_h}\Pi_h^{VEM}\Big(\boldsymbol{\rho},\boldsymbol{\xi}_1,\ldots,\boldsymbol{\xi}_{m_\xi},\mathbf{v}_h\Big),
\end{aligned}\tag{22}$$

where $J(\cdot)$ is the objective function, and $g_V(\boldsymbol{\rho})$ is a global volume constraint with $\mathbf{V}$ and $V_{\max}$ being a vector collecting the volume of each element and the prescribed total volume fraction, respectively.

In the above topology optimization formulation, the vector $\boldsymbol{\rho}$ collects all the design variables associated with the matrix phase, and vectors $\boldsymbol{\xi}_k, k=1,\ldots,m_\xi$ collect the design variables associated with the fiber orientations with m_ξ being the total number of fiber-related design variables in each element. Notice that the value of m_ξ depends on the fiber orientation interpolation scheme chosen and, in general, we have $m_\xi\le m$. With all the design variables defined, the design is parameterized as follows. The design variable vector $\boldsymbol{\rho}$ is responsible for determining the physical density field $\tilde{\boldsymbol{\rho}}$, which characterizes the distribution of materials in the matrix phase. The design variable vectors $\boldsymbol{\xi}_k, k=1,\ldots,m_\xi$, are responsible for determining the fiber orientation fields, $\mathbf{m}_i, i=1,\ldots,m$. A schematic illustration of the design parameterization is provided in Fig. 3(a) and (b) using a problem with 4 candidate fiber orientations. In the sequel, we elaborate the mappings from $\boldsymbol{\rho}$ to $\tilde{\boldsymbol{\rho}}$ and from $\boldsymbol{\xi}_k, k=1,\ldots,m_\xi$ to $\mathbf{m}_i, i=1...,m$, separately.

4.1.1. Mapping from $\boldsymbol{\rho}$ to $\tilde{\boldsymbol{\rho}}$

To map $\boldsymbol{\rho}$ to the physical density $\tilde{\boldsymbol{\rho}}$, we first apply the density filter [49,50] to $\boldsymbol{\rho}$ to obtain an intermediate variable $\bar{\boldsymbol{\rho}}$ via $\bar{\boldsymbol{\rho}}=\mathbf{H}\boldsymbol{\rho}$, where $\mathbf{H}$ is the matrix representation of the density filter, whose expression [51] is given by

$$[\mathbf{H}]_{\ell k}=\frac{\max\Big(0,|E^{(k)}|(1-|\mathbf{x}_\ell^c-\mathbf{x}_k^c|/R)^q\Big)}{\sum_{s\in\mathcal{I}(\ell)}|E^{(s)}|\Big(1-|\mathbf{x}_\ell^c-\mathbf{x}_s^c|/R\Big)^q},\tag{23}$$

where $|E^{(k)}|$ denotes the area of element $E^{(k)}$, $\mathbf{x}_k^c$ is the position vector of centroid of element k in the undeformed configuration, $\mathcal{I}(\ell)$ is the set of indices of element $E^{(k)}$ whose centroid falls within radius R of the centroid of element $E^{(\ell)}$, and q is polynomial exponent of the weight factors.

Then, we apply a smoothed Heaviside function to each component of $\bar{\boldsymbol{\rho}}$ to obtain the physical density field $\tilde{\boldsymbol{\rho}}$, namely,

$$\tilde{\rho}^{(e)}=\frac{\tanh(\beta\rho_0)+\tanh(\beta(\bar{\rho}^{(e)}-\rho_0))}{\tanh(\beta\rho_0)+\tanh(\beta(1-\rho_0))},\tag{24}$$

where β is a parameter that controls the sharpness of the Heaviside function and ρ_0 is the threshold value taken to be 0.5 in this work.

4.1.2. Mapping from $\boldsymbol{\xi}_k$ to $\mathbf{m}_i$

To map $\boldsymbol{\xi}_k, k=1,\ldots,m_\xi$ to $\mathbf{m}_i, i=1,\ldots,m$, we first apply the density filter to $\boldsymbol{\xi}_k$ to ensure their smoothness. The filtered design variables are denoted as $\bar{\boldsymbol{\xi}}_k$, which are computed via $\bar{\boldsymbol{\xi}}_k=\mathbf{H}\boldsymbol{\xi}_k, k=1,\ldots,m_\xi$. We then employ a fiber orientation interpolation scheme to map $\bar{\boldsymbol{\xi}}_k, k=1,\ldots,m_\xi$ to the fiber orientation fields $\mathbf{m}_i, i=1,\ldots,m$.

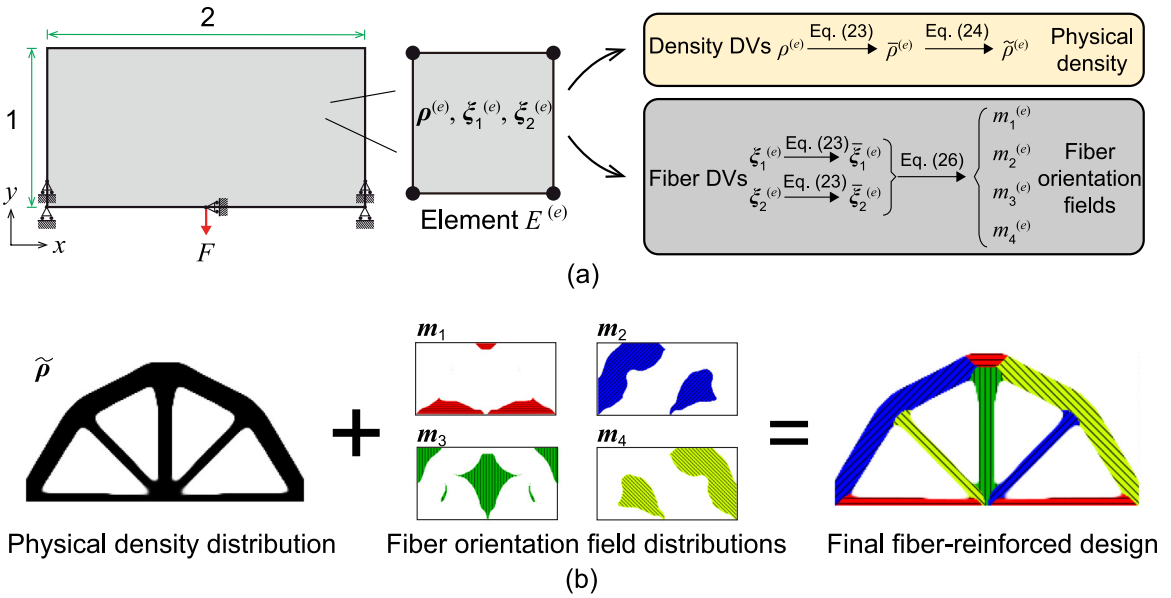


Fig. 3. An illustration of the design parameterization used in this work. (a) A design problem with dimensions, load, and boundary conditions. A set of 4 candidate fiber orientations is considered, i.e., $\{0°, 45°, 90°, -45°\}$. The BCP scheme is adopted in this illustration, leading to a density design variable (DV) $\rho^{(e)}$ and two fiber DVs $\xi_1^{(e)}$ and $\xi_2^{(e)}$ in element $E^{(e)}$ of the mesh. The density and fiber DVs are then mapped to the physical fields following the procedure described in Sections 4.1.1 and 4.1.2 , respectively. (b) The optimized design is composed by a matrix phase described by $\tilde{\rho}$ and four fiber phases represented by $\mathbf{m}_1, \ldots, \mathbf{m}_4$.

In each element, the fiber orientation field $m_i^{(e)}$ varies from 0 to 1, which can be interpreted as the portion of the ith candidate orientation fiber among all fibers in element e. Therefore, to ensure the physical meaning, the fiber orientation fields need to satisfy the following physical constraint:

$$\sum_{i=1}^{m} m_i^{(e)} \leq 1, \quad e = 1, \ldots, M. \tag{25}$$

In this work, the above constraint is ensured via the fiber orientation schemes. In particular, we consider the following two fiber orientation interpolation schemes: the BCP scheme [15] and the hypercube-to-simplex projection (HSP) scheme [19], both of which are discussed as follows.

Bi-value coding parameterization (BCP) scheme. In the BCP scheme, the fiber orientation field $m_i^{(e)}$ of each element is interpolated as [15]

$$m_i^{(e)} = \frac{1}{2^{m_\xi}} \prod_{k=1}^{m_\xi} \left(1 + s_{ik} \bar{\xi}_k^{(e)}\right), \tag{26}$$

where

$$s_{ik} = \begin{cases} 1 & i \in [1, 2^{k-1}] \\ -1 & i \in [2^{k-1}+1, 2^k] \\ s_{\zeta k} & i \in [2^k+1, 2^{m_\xi}], \quad \text{where } \zeta = 2^{\lceil \log_2 k \rceil} + 1 - i, \end{cases} \tag{27}$$

with $\xi_{\min} = -1$ and $\xi_{\max} = 1$. In this BCP interpolation scheme, we require a total number of $m_\xi = \lceil \log_2 m \rceil$ design variables per element, where $\lceil \cdot \rceil$ denotes the ceil function. The expression of the derivative of $m_i^{(e)}$ with respect to $\bar{\xi}_k^{(e)}$ is

$$\frac{\partial m_i^{(e)}}{\partial \bar{\xi}_k^{(e)}} = \frac{s_{ik}}{2^{m_\xi}} \prod_{\substack{j=1 \\ j \neq k}}^{m_\xi} \left(1 + s_{ij} \bar{\xi}_j^{(e)}\right), \tag{28}$$

where the coefficients s_{ij} are given by (27).

Hypercube-to-simplex projection (HSP) scheme. The HSP scheme is proposed in [19] with the idea of mapping the design variables from a m-dimensional hypercube to a m-dimensional simplex so that the unity constraint $\sum_{i=1}^{m} m_i^{(e)} \leq 1$ is implicitly satisfied. In this scheme, the fiber orientation field $m_i^{(e)}$ of each element is interpolated as

$$m_i^{(e)} = \sum_{k=1}^{2^m} b_{ik} \left\{ (-1)^{(m+\sum_{\ell=1}^{m} c_{k\ell})} \prod_{n=1}^{m} \left(\bar{\xi}_n^{(e)} + c_{kn} - 1 \right) \right\}, \tag{29}$$

where $c_{kn} \in \{0, 1\}$ is the nth component of the coordinate of the kth vertex of a unit hypercube in a m dimensional space; and the coefficient b_{ik} is defined as

$$b_{ik} = \begin{cases} \frac{c_{ik}}{\sum_\ell c_{i\ell}} & \text{if } \sum_\ell c_{i\ell} \geq 1 \\ 0 & \text{Otherwise.} \end{cases} \tag{30}$$

The expression of the derivative of the $m_i^{(e)}$ with respect to $\bar{\xi}_k^{(e)}$ is given by

$$\frac{\partial m_i^{(e)}}{\partial \bar{\xi}_k^{(e)}} = \sum_{j=1}^{2^m} b_{ij} \left\{ (-1)^{(m+\sum_{\ell=1}^{m} c_{j\ell})} \prod_{\substack{n=1 \\ n \neq k}}^{m} \left(\bar{\xi}_n^{(e)} + c_{jn} - 1 \right) \right\}, \tag{31}$$

where the coefficient b_{ij} is given by expression (30).

Remark 1. Comparing the two fiber interpolation schemes adopted, we notice that the BCP scheme needs fewer design variables as compared to the HSP scheme, i.e., $m_\xi = \lceil \log_2 m \rceil$ v.s. $m_\xi = m$, and the difference becomes larger as the number of candidate orientations increases. As demonstrated in Section 5, the update of the design variables associated with the fiber orientation fields can be decoupled, which means the HSP scheme can be made as efficient as BCP scheme in the topology optimization framework (22).

Remark 2. In addition to the BCP and the HSP schemes adopted in this work, any interpolation scheme which satisfies the property of $\sum_{i=1}^{m} m_i^{(e)} \leq 1$, such as the NDFO [17,21] and DMO [13], could be potentially adopted in the proposed framework as well.

4.2. Anisotropic material interpolation for fiber-reinforced elastomers

We now propose an anisotropic material interpolation scheme for the composite material, i.e., the interpolation of the stored-energy function of the fiber-reinforced hyperelastic structures from the physical density and fiber orientation fields. For a given element e with physical density, $\tilde{\rho}^{(e)}$, and fiber orientation fields, $m_1^{(e)}, \ldots, m_m^{(e)}$, we define the stored-energy function of that element as

$$\Psi^{\rho,m}\left(\rho^e, \xi_1^{(e)}, \ldots, \xi_{m_\xi}^{(e)}, \mathbf{v}\right) = w^{(e),\rho} \left[\Psi_{\mathrm{M}}\left(I_1(\mathbf{v}), I_2(\mathbf{v}), J(\mathbf{v})\right) + \sum_{i=1}^{m} w_i^{(e),m} \Psi_{\mathrm{F}}\left(I_4(\mathbf{a}_0^i, \mathbf{v})\right) \right], \tag{32}$$

where $\mathbf{a}_0^i$ is the orientation associated with ith candidate fiber family in the undeformed configuration, $w^{(e),\rho}$ and $w_i^{(e),m}$ are the weights associated with the density field and the ith fiber orientation field in the eth element, respectively.

The density weight $w^{(e),\rho}$ is associated with the stored-energy functions of both matrix and fiber, and is determined through the Solid Isotropic Material with Penalization (SIMP) method [52,53], namely,

$$w^{(e),\rho} = \varepsilon + \left(1 - \varepsilon\right)\left(\tilde{\rho}^{(e)}\right)^{p_\rho}, \tag{33}$$

where p_ρ is the SIMP penalization parameter associated with the density and ε is the Ersatz parameter taking to be $\varepsilon = 10^{-4}$ throughout this work. The weights $w_i^{(e),m}, i = 1, \ldots, m$, associated with the ith fiber orientation field are interpolated using the SIMP method as

$$w_i^{(e),m} = \left(m_i^{(e)}\right)^{p_m}, \quad i = 1, \ldots, m, \tag{34}$$

where p_m is the SIMP penalization parameter associated with fiber orientation fields.

4.3. Energy interpolation and VEM approximation of potential energy function

In topology optimization problems involving large deformations, low density elements can undergo excessive deformations which may cause convergence difficulty in solving the nonlinear state equation. Built upon the anisotropic material interpolation scheme proposed in the preceding subsection, this subsection introduces an energy interpolation scheme to address the numerical challenge associated with the excessive deformations of low density elements [54].

The main idea of the energy interpolation scheme is to use the original stored-energy function $\Psi^{\rho,m}$ for elements with relatively high densities and model the low-density elements (void) as linear elastic materials [54]. Thus, we adopt an energy interpolation parameter, $\gamma^{(e)}$, as

$$\gamma^{(e)} = \frac{\tanh(\beta_\gamma \rho_\gamma) + \tanh\Big(\beta_\gamma (w^{(e),\rho} - \rho_\gamma)\Big)}{\tanh(\beta_\gamma \rho_\gamma) + \tanh\Big(\beta_\gamma (1 - \rho_\gamma)\Big)}, \tag{35}$$

where we recall that $w^{(e),\rho}$ is the weight associated with the density field defined in the anisotropic material interpolation for element e, ρ_γ is the threshold, and β_γ is a parameter that controls the sharpness of the transition zone. Throughout this work, we choose $\beta_\gamma = 500$, which ensures that the transition zone is sharp, and take $\rho_\gamma = 0.005$. According to the definition of the energy interpolation scheme, for those elements whose physical densities are above ρ_γ, we have $\gamma^{(e)} = 1$; otherwise, we have $\gamma^{(e)} = 0$. With $\gamma^{(e)}$, the final stored-energy function is given by [54]

$$\begin{aligned}\Psi_\gamma^{\rho,m}(\rho^{(e)}, \xi_1^{(e)}, \ldots, \xi_{m_\xi}^{(e)}, \gamma^{(e)}, \mathbf{v}) &= \Psi^{\rho,m}(\rho^{(e)}, \xi_1^{(e)}, \ldots, \xi_{m_\xi}^{(e)}, \gamma^{(e)}\mathbf{v}) - \Psi_L^{\rho}(\rho^{(e)}, \gamma^{(e)}\mathbf{v}) + \Psi_L^{\rho}(\rho^{(e)}, \mathbf{v}) \\ &= \Psi^{\rho,m}(\rho^{(e)}, \xi_1^{(e)}, \ldots, \xi_{m_\xi}^{(e)}, \gamma^{(e)}\mathbf{v}) + \Big(1 - (\gamma^{(e)})^2\Big)\Psi_L^{\rho}(\rho^{(e)}, \mathbf{v}),\end{aligned} \tag{36}$$

where Ψ_L^ρ stands for the stored-energy function for isotropic linear elastic material taking the form

$$\Psi_L^{\rho}(\rho^{(e)}, \mathbf{v}) = w^{(e),\rho}\Big[\mu \boldsymbol{\varepsilon}(\mathbf{v}) : \boldsymbol{\varepsilon}(\mathbf{v}) + \frac{3\kappa - 4\mu}{6}\Big(\mathrm{tr}\boldsymbol{\varepsilon}(\mathbf{v})\Big)\Big], \tag{37}$$

where μ and κ are the shear and bulk moduli of the matrix phase, respectively, and $\boldsymbol{\varepsilon}(\mathbf{v}) = 1/2(\nabla\mathbf{v} + (\nabla\mathbf{v})^T)$ is the linearized strain tensor.

The stored-energy function $\Psi_\gamma^{\rho,m}$ is used to form the total potential energy function of the fiber-reinforced composites. Plugging it into the VEM approximation of the stored-energy function in (14), we obtain the final VEM approximation of the state equation as

$$\begin{aligned}&\Pi_h^{VEM}\Big(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \boldsymbol{\gamma}, \mathbf{v}_h\Big) \\ &= \sum_{e=1}^{M}\Bigg\{|E^{(e)}|\Big[\Psi_\gamma^{\rho,m}\Big(\rho^{(e)}, \xi_1^{(e)}, \ldots, \xi_{m_\xi}^{(e)}, \gamma^{(e)}\boldsymbol{\Pi}_{E^{(e)}}^{\nabla}\mathbf{v}_h\Big) + \Big(1 - (\gamma^{(e)})^2\Big)\Psi_L^{\rho}\Big(\rho^{(e)}, \boldsymbol{\Pi}_{E^{(e)}}^{\nabla}\mathbf{v}_h\Big)\Big] \\ &\quad + \frac{w^{(e),\rho}}{2}\alpha_{E^{(e)}}^{\gamma}(\gamma^{(e)}, \boldsymbol{\Pi}_{E^{(e)}}^{\nabla}\mathbf{v}_h)S_h^{E^{(e)}}(\mathbf{v}_h - \boldsymbol{\Pi}_{E^{(e)}}^{\nabla}\mathbf{v}_h, \mathbf{v}_h - \boldsymbol{\Pi}_{E^{(e)}}^{\nabla}\mathbf{v}_h)\Bigg\} \\ &\quad - \langle \mathbf{t}, \mathbf{v}_h\rangle_h,\end{aligned} \tag{38}$$

where $\alpha_{E^{(e)}}^{\gamma}$ is the VEM stability parameter obtained from the energy interpolation, which is

$$\alpha_{E^{(e)}}^{\gamma} = \big(\gamma^{(e)}\big)^2\alpha_{E^{(e)}}\big(\gamma^{(e)}\boldsymbol{\Pi}_{E^{(e)}}^{\nabla}\big) + \big(1 - (\gamma^{(e)})^2\big)\Big(\frac{1}{d}\kappa + (1 + \frac{1}{3d})\mu\Big), \tag{39}$$

with μ and κ are the shear and bulk moduli of the matrix material.

In our implementation, the above total potential energy Π_h^{VEM} is linearized with respect to the nodal displacement vector via $\partial\Pi_h^{VEM}/\partial\mathbf{v}_h = \mathbf{0}$ [28,47], which gives the following nonlinear system of equations:

$$\mathbf{R}\Big(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \boldsymbol{\gamma}, \mathbf{v}_h\Big) = \mathbf{F}_{\text{int}}\Big(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \boldsymbol{\gamma}, \mathbf{v}_h\Big) - \mathbf{F}_{\text{ext}} = \mathbf{0}, \tag{40}$$

where $\mathbf{R}$ is the global residual vector; $\mathbf{F}_{\text{int}}$ and $\mathbf{F}_{\text{ext}}$ stand for the global internal and external force vectors, respectively. The Newton's method is then employed to solve for the equilibrating displacement $\mathbf{u}_h$.

5. A decoupled fiber-matrix design variable update scheme

Traditionally, the design variables $\boldsymbol{\rho}$ and $\boldsymbol{\xi}_i, i = 1, \ldots, m_\xi$, in the optimization formulation (22) need to be updated in a coupled fashion. Inspired by the ZPR scheme [22], we derive that the update of $\boldsymbol{\rho}$ and each $\boldsymbol{\xi}_i$ should in fact be performed in a decoupled manner analytically. Furthermore, we show that an explicit update formula for the design variables $\boldsymbol{\xi}_i, i = 1, \ldots, m_\xi$ can be derived.

We first denote the design variables at the optimization step k as $\boldsymbol{\rho}_k$ and $\boldsymbol{\xi}_{i,k}, i = 1, \ldots, m_\xi$. We then adopt the explicit convex approximation used in the MMA [55] and construct an explicit convex sub-problem of the form:

$$\min_{\boldsymbol{\rho},\boldsymbol{\xi}_i \in \mathbb{R}^M} \sum_{e=1}^{M} \left\{ \left[\frac{P_{J,k}^{(e),\rho}}{U_{J,k}^{(e),\rho} - \rho^{(e)}} + \frac{Q_{J,k}^{(e),\rho}}{\rho^{(e)} - L_{J,k}^{(e),\rho}} \right] + \sum_{j=1}^{m_\xi} \left[\frac{P_{J,k}^{(e),\xi_j}}{U_{J,k}^{(e),\xi_j} - \xi_j^{(e)}} + \frac{Q_{J,k}^{(e),\xi_j}}{\xi_j^{(e)} - L_{J,k}^{(j),\xi_j}} \right] \right\} \tag{41}$$

$$\text{s.t.} \quad g_V^k(\boldsymbol{\rho}_k) + \sum_{e=1}^{M} \left[\frac{P_{g,k}^{(e),\rho}}{U_{g,k}^{(e),\rho} - \rho^{(e)}} + \frac{Q_{g,k}^{(e),\rho}}{\rho^{(e)} - L_{g,k}^{(e),\rho}} \right] \leq 0 \tag{42}$$

$$\rho_L^{(e)} \leq \rho^{(e)} \leq \rho_U^{(e)} \qquad e = 1, \ldots, M \tag{43}$$

$$\xi_{i,L}^{(e)} \leq \xi_i^{(e)} \leq \xi_{i,U}^{(e)} \qquad e = 1, \ldots, M \text{ and } i = 1, \ldots, m_\xi \tag{44}$$

where $U_{J,k}^{(e),\rho}$ and $L_{J,k}^{(e),\rho}$ are the upper and lower asymptotes for the objective function approximation associated with $\rho^{(e)}$, which are consistently updated during optimization [55]; $\rho_L^{(e)}$ and $\rho_U^{(e)}$ are the lower and upper bounds of $\rho^{(e)}$, respectively; and $P_{J,k}^{(e),\rho}$ and $Q_{J,k}^{(e),\rho}$ are given by

$$P_{J,k}^{(e),\rho} = \begin{cases} \left(U_{J,k}^{(e),\rho} - \rho_k^{(e)} \right)^2 \frac{\partial J}{\partial \rho^{(e)}} & \text{if} \quad \frac{\partial J}{\partial \rho^{(e)}} > 0 \\ 0 & \text{if} \quad \frac{\partial J}{\partial \rho^{(e)}} \leq 0 \end{cases} \tag{45}$$

$$Q_{J,k}^{(e),\rho} = \begin{cases} 0 & \text{if} \quad \frac{\partial J}{\partial \rho^{(e)}} \geq 0 \\ -\left(\rho_k^{(e)} - L_{J,k}^{(e),\rho} \right)^2 \frac{\partial J}{\partial \rho^{(e)}} & \text{if} \quad \frac{\partial J}{\partial \rho^{(e)}} < 0 \end{cases} \tag{46}$$

with $\partial J / \partial \rho^{(e)}$ being evaluated at the kth optimization step. Notice that, in the above discussion, we only define the notations for the parameters associated with the design variable $\rho^{(e)}$, the ones associated with design variables $\xi_i^{(e)}$ are defined in the same way and therefore are not listed here for brevity. For the convex approximation of the volume constraint function, the relevant parameters also follow the same definitions, except for $g_V^k(\boldsymbol{\rho}_k)$, which is given as [55]

$$g_V^k(\boldsymbol{\rho}_k) = g_V(\boldsymbol{\rho}_k) - \sum_{e=1}^{M} \left[\frac{P_{g,k}^{(e),\rho}}{U_{g,k}^{(e),\rho} - \rho_k^{(e)}} + \frac{Q_{g,k}^{(e),\rho}}{\rho_k^{(e)} - L_{g,k}^{(e),\rho}} \right]. \tag{47}$$

Given the subproblem (41), we can show that its solution can be obtained equivalently via a series of decoupled subproblems, each for one design variable set. For the density design variables $\boldsymbol{\rho}$, we have the decoupled subproblem of the form

$$\begin{aligned} &\min_{\boldsymbol{\rho} \in \mathbb{R}^M} \sum_{e=1}^{M} \left[\frac{P_{J,k}^{(e),\rho}}{U_{J,k}^{(e),\rho} - \hat{\rho}^{(e)}} + \frac{Q_{J,k}^{(e),\rho}}{\hat{\rho}^{(e)} - L_{J,k}^{(e),\rho}} \right] \\ &\text{s.t.} \quad g_V^k(\boldsymbol{\rho}_k) + \sum_{e=1}^{M} \left[\frac{P_{g,k}^{(e),\rho}}{U_{g,k}^{(e),\rho} - \hat{\rho}^{(e)}} + \frac{Q_{g,k}^{(e),\rho}}{\hat{\rho}^{(e)} - L_{g,k}^{(e),\rho}} \right] \leq 0 \\ &\qquad \hat{\rho}_L^{(e)} \leq \hat{\rho}^{(e)} \leq \hat{\rho}_U^{(e)} \qquad e = 1, \ldots, M. \end{aligned} \tag{48}$$

For fiber orientation design variables $\boldsymbol{\xi}_i, i = 1, \ldots, m_\xi$, we have the decoupled subproblem of form

$$\min_{\boldsymbol{\xi}_i \in \mathbb{R}^M} \sum_{e=1}^{M} \left[\frac{P_{J,k}^{(e),\xi_i}}{U_{J,k}^{(e),\xi_i} - \xi_i^{(e)}} + \frac{Q_{J,k}^{(e),\xi_i}}{\xi_i^{(e)} - L_{J,k}^{(e),\xi_i}} \right] \quad \text{s.t.} \quad \xi_{i,L}^{(e)} \leq \xi_i^{(e)} \leq \xi_{i,U}^{(e)} \qquad e = 1, \ldots, M. \tag{49}$$

In practice, we can solve the subproblem (48) using the MMA [55] to obtain the design variables update for density $\boldsymbol{\rho}$. For fiber orientation design variables $\boldsymbol{\xi}_i, i = 1, \ldots, m_\xi$, we notice that the subproblems (49) only have

box constraints. Therefore, explicit expressions can be directly derived from Karush–Kuhn–Tucker (KKT) conditions of (49), which is

$$\xi_{i,k+1}^{(e)} = \begin{cases} \xi_{i,L}^{(e)} & \text{if } \xi_{i,L}^{(e)} < \xi_i^{(e),*} \\ \xi_i^{(e),*} & \text{if } \xi_{i,L}^{(e)} \leq \xi_i^{(e),*} \leq \xi_{i,U}^{(e)} \\ \xi_{i,U}^{(e)} & \text{if } \xi_i^{(e),*} > \xi_{i,U}^{(e)} \end{cases} \tag{50}$$

where

$$\xi_i^{(e),*} = \frac{P_{J,k}^{(e),\xi_i} L_{J,k}^{(e),\xi_i} - Q_{J,k}^{(e),\xi_i} U_{J,k}^{(e),\xi_i} + \sqrt{P_{J,k}^{(e),\xi_i} Q_{J,k}^{(e),\xi_i} \left(L_{J,k}^{(e),\xi_i} - U_{J,k}^{(e),\xi_i}\right)^2}}{P_{J,k}^{(e),\xi_i} - Q_{J,k}^{(e),\xi_i}}. \tag{51}$$

Remark 1. With the decoupled subproblems, we can now perform independent update for $\boldsymbol{\rho}$ and for each individual $\boldsymbol{\xi}_i, i = 1, \ldots, m_\xi$, in parallel. This proposed decoupled update scheme greatly reduces the associated computational cost, especially for cases where large numbers of design variables and candidate fiber orientations are considered.

Remark 2. Although the explicit convex approximation used in MMA [55] is considered in this paper, other convex approximation forms could be potentially adopted and the corresponding decoupled update formula can be derived following the procedures introduced in this section.

6. Design examples

In this section, we present three design examples that demonstrate the effectiveness of the proposed framework in simultaneously optimizing both the material distribution of the matrix phase and the fiber orientations in the reinforcement phase.

We consider three objective functions to showcase the applicability of the proposed framework to a wide range of problems. The first example considers the classic end-compliance minimization problem [56] with the objective function J of the following form,

$$J(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \mathbf{u}_h) = \mathbf{F}_{ext}^T \mathbf{U}, \tag{52}$$

where $\mathbf{U}$ is the global nodal displacement vector collecting the nodal values of $\mathbf{u}_h$, and $\mathbf{F}_{ext}$ is the external force vector. In the second example, our goal is to maximize the potential energy of the systems in the equilibrium state. The corresponding objective function has the form of [25]

$$J(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \mathbf{u}_h) = -\Pi_h^{VEM}\left(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \mathbf{u}_h\right), \tag{53}$$

where $\mathbf{u}_h$ is the displacement field at the equilibrium state. An advantage of this objective function is that the solution of the adjoint system is not needed when performing the sensitivity analysis. In the last design example, we consider a compliant mechanism optimization with the objective function being the form

$$J(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \mathbf{u}_h) = \mathbf{L}^T \mathbf{U}, \tag{54}$$

where $\mathbf{U}$ is the nodal displacement vector of $\mathbf{u}_h$ and $\mathbf{L}$ is a constant vector whose components are 1 at the output DOFs and 0 elsewhere.

For all the design examples, the density design variable $\boldsymbol{\rho}$ and fiber orientation design variables $\boldsymbol{\xi}_i, i = 1, \ldots, m_\xi$, are filtered using the density filter with linear (i.e., $q = 1$ in (23)) and quadratic (i.e., $q = 2$ in (23)) kernels, respectively. Unless otherwise stated, the decoupled design variable update scheme introduced in Section 5 is used. We employ uniform initial guesses for both density and fiber orientation design variables. For the density design variables, the initial guess is taken as $\rho^{(e)} = V_{\max}$, where $V_{\max}$ is the allowable volume fraction. For the fiber orientation design variables, the initial guess is chosen as $\xi_i^{(e)} = 0, i = 1, \ldots, m_\xi$, if the BCP scheme is used; and $\xi_i^{(e)} = 1, i = 1, \ldots, m_\xi$, if the HSP scheme is used. For both cases, the initial fiber orientation design variables correspond to designs where all the fiber phases are present with equal amount (i.e., $1/m$, where m is the total number of candidate fiber orientations).

In the final designs, the fiber orientation of each element in the meshes will converge to one of the pre-defined directions, and elements with different optimized fiber orientation will be plotted with different colors (both in the

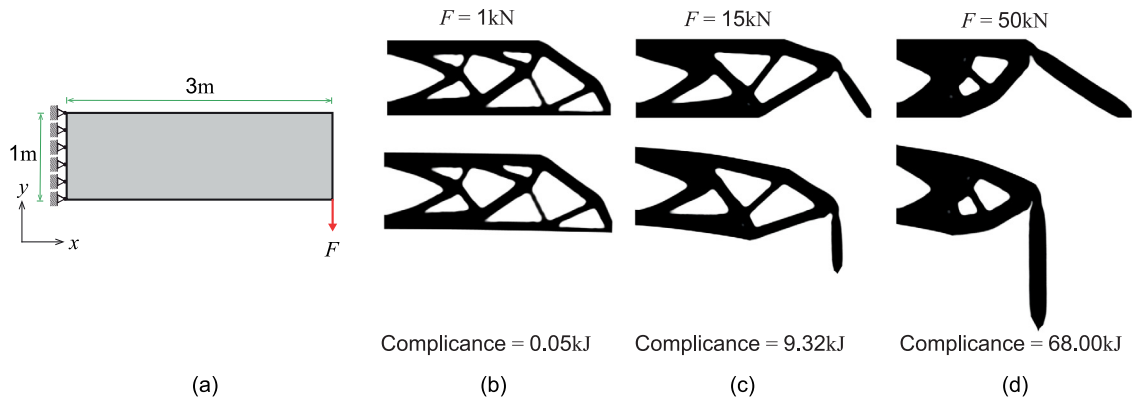


Fig. 4. (a) Design domain, load, and boundary conditions of Example 1. (b)–(d): Optimized designs obtained by load levels $F = 1$ kN, $F = 15$ kN, and $F = 50$ kN, respectively. The optimized designs are shown in both undeformed (top row) and deformed (bottom row) configurations.

Table 1
A list of parameters considered in Example 1.

Load levels F (kN)	(i) 1; (ii) 15; and (iii) 50
Volume fraction upper bound $V_{\max}$	50%
Maximum optimization step	300
Matrix Young's modulus E_0 (kPa) and Poisson's ratio ν	3,000 and 0.35
Fiber moduli E_t (kPa) and E_c (kPa)	6,000 and 3,000

undeformed and deformed configurations). In addition, we will plot solid lines to better and more explicitly visualize the distributions of optimized fiber orientations in the final designs in their undeformed configurations. Those lines and their geometries (e.g., spacing and length), however, are for schematic illustration purposes and not related to the actual underlying meshes.

6.1. Example 1: end-compliance minimization

The design domain, load, and boundary conditions of the first example are shown in Fig. 4(a). The design domain is discretized using an initial mesh with an 108×36 structured grid, which is then adaptively refined and coarsened during the optimization process [24,25]. The adaptive refinement starts at optimization step 10 and is performed every 10 optimization steps afterwards. We initially allow for 1 refinement level and, after optimization step 190, 2 refinement levels are allowed. The adaptive coarsening starts at optimization step 12 and is performed every 3 optimization steps afterwards. The modified Generalized Displacement Control [57] is adopted with a tolerance of 10^{-6}.

We adopt continuation for parameters p_ρ, p_m, β and R. The parameter p_ρ is initially taken as 1 and is increased every optimization step by 0.2 until it reaches $p_\rho = 3$. The parameter p_m is initially taken as 1 and, starting from optimization step 60, is increased every 4 optimization steps by 0.1 until $p_m = 3$ is reached. The filter radii for both the density and fiber orientation fields are taken as $R = 0.1$ m and, starting from optimization step 190, the filter radius for the fiber orientation fields is reduced every 10 optimization steps by $\Delta R = -0.009$ m until it reaches $R_{\min} = 0.01$ m. Finally, the β parameter for the Heaviside filter starts as $\beta = 1$ and is doubled every 30 steps starting from optimization step 120 until it reaches $\beta = 64$. Other design parameters and the arrangement of candidate orientations in each fiber set are summarized in Tables 1 and 2, respectively.

The optimized designs for three load levels ($F = 1$ kN, $F = 15$ kN, and $F = 50$ kN) with corresponding compliance values obtained by considering only the matrix material are shown in Fig. 4(b)–(d) in both undeformed and deformed configurations. We also consider these three load levels and both the BCP and HSP fiber interpolation schemes (see Section 4.1.2) for four candidate fiber sets (see Table 2) in fiber-reinforced designs. In Figs. 5 and 6, we plot the optimized designs (in their undeformed configurations with visualized fiber reinforcement) obtained

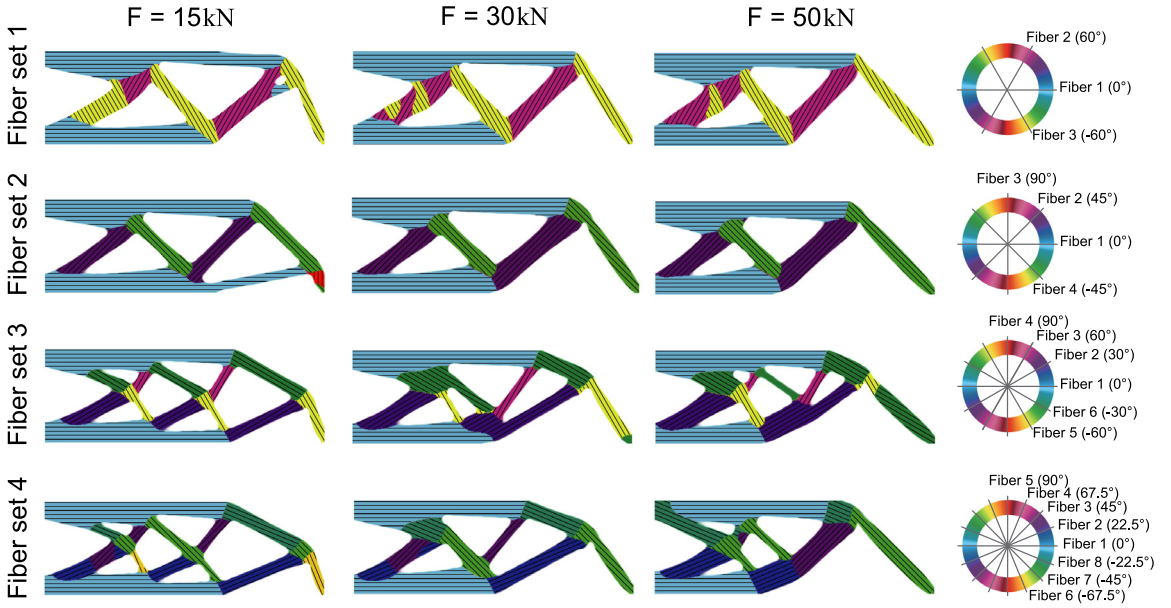


Fig. 5. Optimized designs obtained by the BCP fiber interpolation scheme considering four fiber sets and three load levels. The designs are presented in their undeformed configurations with fiber reinforcements explicitly plotted.

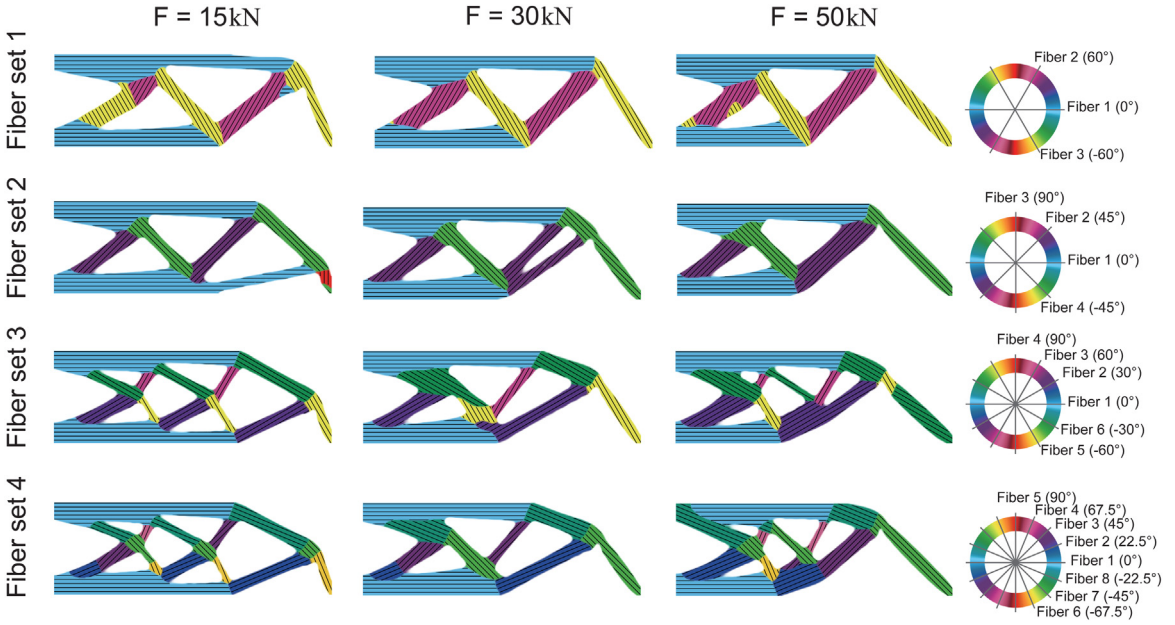


Fig. 6. Optimized designs obtained by the HSP fiber interpolation scheme considering four fiber sets and three load levels. The designs are presented in their undeformed configurations with fiber reinforcements explicitly plotted.

by the BCP and HSP schemes, respectively. In addition, Figs. 7–9 show comparisons of the final compliance values and corresponding designs (in their deformed configurations) obtained by different numbers of candidate fiber orientations for load levels $F = 1$ kN, $F = 15$ kN, and $F = 50$ kN, respectively. Moreover, Table 3 provides a comparison of the objective function values of the optimized designs and those of the initial designs (with

Table 2
Fiber orientations in each candidate fiber set considered in the first design example.

Fiber set	Number of orientations	Fiber orientations
1	3	{0°, 60°, −60°}
2	4	{0°, 45°, 90°, −45°}
3	6	{0°, 30°, 60°, 90°, −60°, −30°}
4	8	{0°, 22.5°, 45°, 67.5°, 90°, −67.5°, −45°, −22.5°}

Table 3
Comparison of the compliance values (unit: kJ) of the optimized designs with those of the initial designs for all the design cases. Only results obtained from the HSP fiber interpolation scheme are presented, those from the BCP scheme have the same trends and thus are not listed.

Fiber set		1	2	3	4
$F = 1$ kN	Initial design obj.	0.048	0.048	0.049	0.048
	Optimized design obj.	0.025	0.024	0.024	0.024
$F = 15$ kN	Initial design obj.	9.27	9.28	9.27	9.26
	Optimized design obj.	5.41	5.09	4.91	4.91
$F = 50$ kN	Initial design obj.	72.37	72.33	72.32	72.32
	Optimized design obj.	45.10	43.67	43.86	43.07

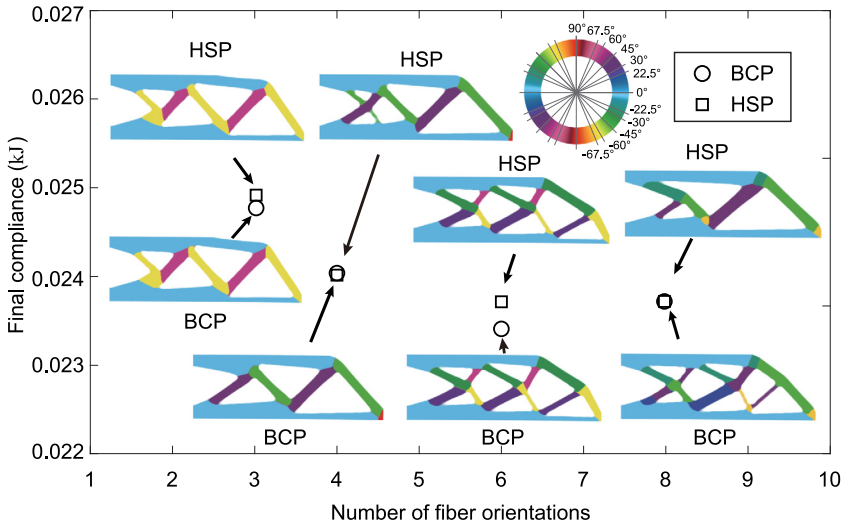


Fig. 7. Comparisons of the final compliance values and corresponding designs (in deformed configurations) obtained by considering 3, 4, 6, and 8 candidate fiber orientations for the load level $F = 1$ kN.

initial guess of design variables and initial parameters). For fair comparison, the objective functions of the initial designs are evaluated on the final meshes obtained after the adaptive mesh refinement and coarsening procedure. The comparison showcases that the optimization produces designs that have improved performance than the initial designs. We note that Table 3 only presents comparisons for the HSP fiber interpolation scheme, those for the BCP scheme have the same trends and thus are not listed.

Several observations can be made. First, the comparison of Fig. 4 with Figs. 7–9 demonstrates that, as expected, adding fiber reinforcement to the matrix phase can greatly reduce the compliance of the optimized designs, especially for larger load levels (e.g., $F = 50$ kN). Moreover, with both the BCP and HSP fiber orientation interpolation schemes, it is a general trend (with a few exception cases) that increasing the number of candidate fiber orientations leads to optimized designs with smaller compliance values as results of the enlarged design spaces. However, we also note that this trend could be problem-dependent and may not immediately generalize to other problems due to the non-convexity of the optimization formulation. Second, the number and the distribution of fiber orientations in the

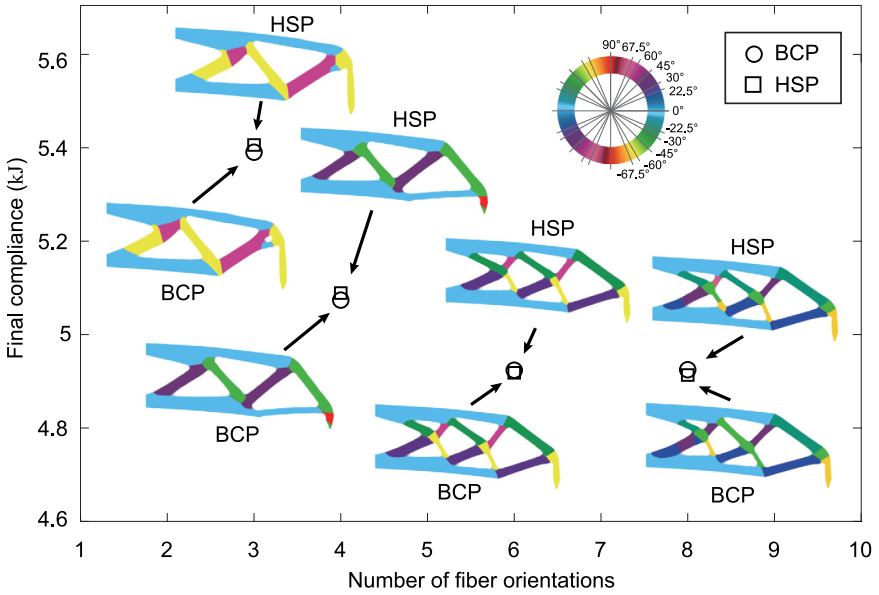


Fig. 8. Comparisons of the final compliance values and corresponding designs (in deformed configurations) obtained by considering 3, 4, 6, and 8 candidate fiber orientations for the load level $F = 15$ kN.

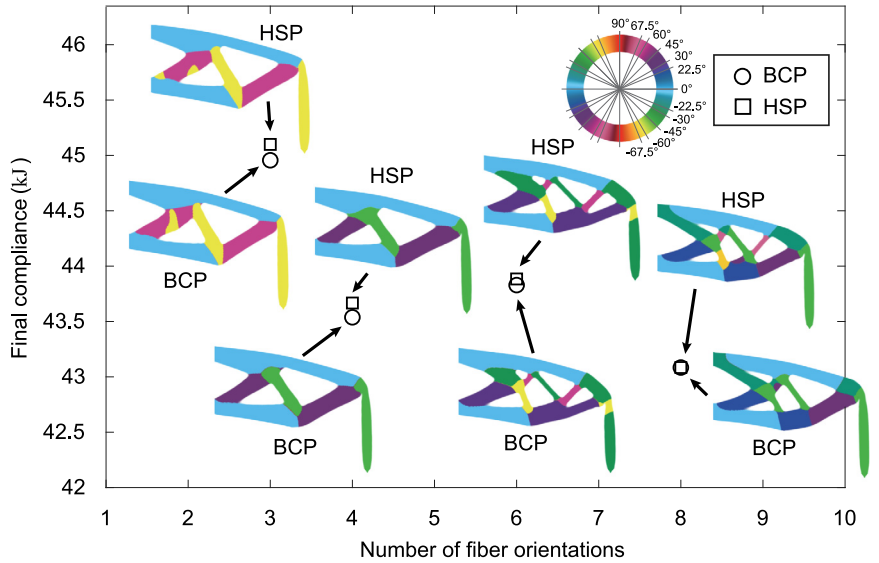


Fig. 9. Comparisons of the final compliance values and corresponding designs (in deformed configurations) obtained by considering 3, 4, 6, and 8 candidate fiber orientations for the load level $F = 50$ kN.

sets have apparent influence on the layout of the optimized designs and, in particular, the distribution of material in the matrix phase (as shown in Figs. 5 and 6). In general, the structural members in the optimized designs tend to be closely aligned with the associated fiber orientations. For example, the designs obtained by considering 4 candidate fiber orientations (i.e., fiber set 2) are distinct from those obtained by considering 8 candidate fiber orientations (i.e., fiber set 4). Compared to the latter ones (with 8 fiber orientations), the structural members in the former ones (with 4 fiber orientations) are oriented in relatively limited directions — most of them are close to $0°, 45°$ and $-45°$, which are in fiber orientation set 2. Third, through a closer comparison of the designs obtained from the BCP and HSP fiber orientation interpolation schemes, we see that the two schemes lead to designs with similar

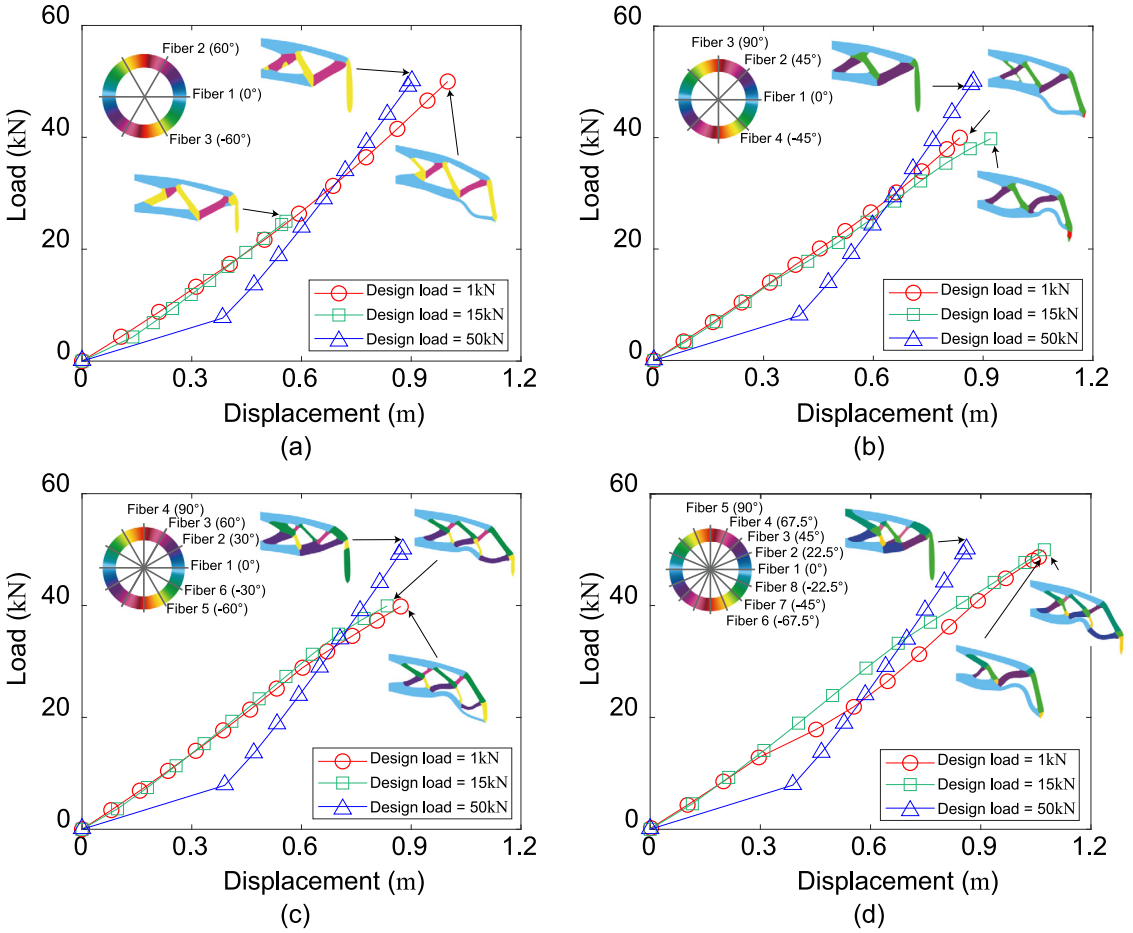


Fig. 10. The load–displacement curves of the optimized designs obtained using the HSP scheme for (a) fiber set 1, (b) fiber set 2, (c) fiber set 3, and (d) fiber set 4. The optimized designs obtained by the three load levels are loaded until either $F = 50$ kN or the solver convergence failure is reached.

distributions of fiber orientations when the number of candidate fiber orientations is relatively small (e.g., 3 and 4). When the number of orientations increases (e.g., to 6 and 8), the two fiber orientation interpolation schemes could lead to designs with different material and fiber orientation distributions in the middle regions of the design domain, while the final compliance values of those designs are still comparable, reflecting the non-convex nature of the optimization problem.

Furthermore, to demonstrate the influence and importance of considering nonlinearity and large deformations in the optimization, we perform load–displacement analyses on the optimized designs obtained using the HSP scheme. The ones obtained using BCP scheme have similar behaviors and thus are not shown for brevity. In Fig. 10(a)–(d), we load the optimized designs obtained by considering the three load levels until either $F = 50$ kN or the solver convergence failure is reached, and plot their corresponding load–displacement curves. For all the fiber sets, it is observed that, although the designs optimized considering smaller load levels (i.e., $F = 1$ kN and $F = 15$ kN) have load–displacement curves that are initially stiffer than the ones optimized considering $F = 50$ kN, they experience local buckling as load F increases beyond their design magnitudes and then become softer than the optimized designs obtained under $F = 50$ kN. Qualitatively comparing to the optimized designs obtained under smaller load levels, the improved robustness of the optimized designs obtained under $F = 50$ kN is a consequence of having thicker members in compression. Additionally, Fig. 11(a)–(d) plots the histograms of the principal stretches of each element in the designs optimized under $F = 50$ kN, for fiber sets 1 to 4, respectively. Only those elements

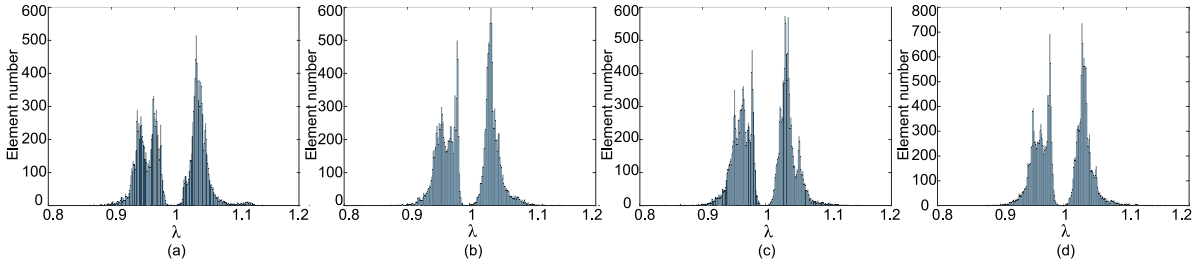


Fig. 11. The principal stretch levels within each elements are quantified for the optimized designs obtained by $F = 50$ kN; (a)–(d): fiber set 1 to fiber set 4. Only those elements whose optimized physical densities $\tilde{\rho}^{(e)}$ greater than 0.5 are counted.

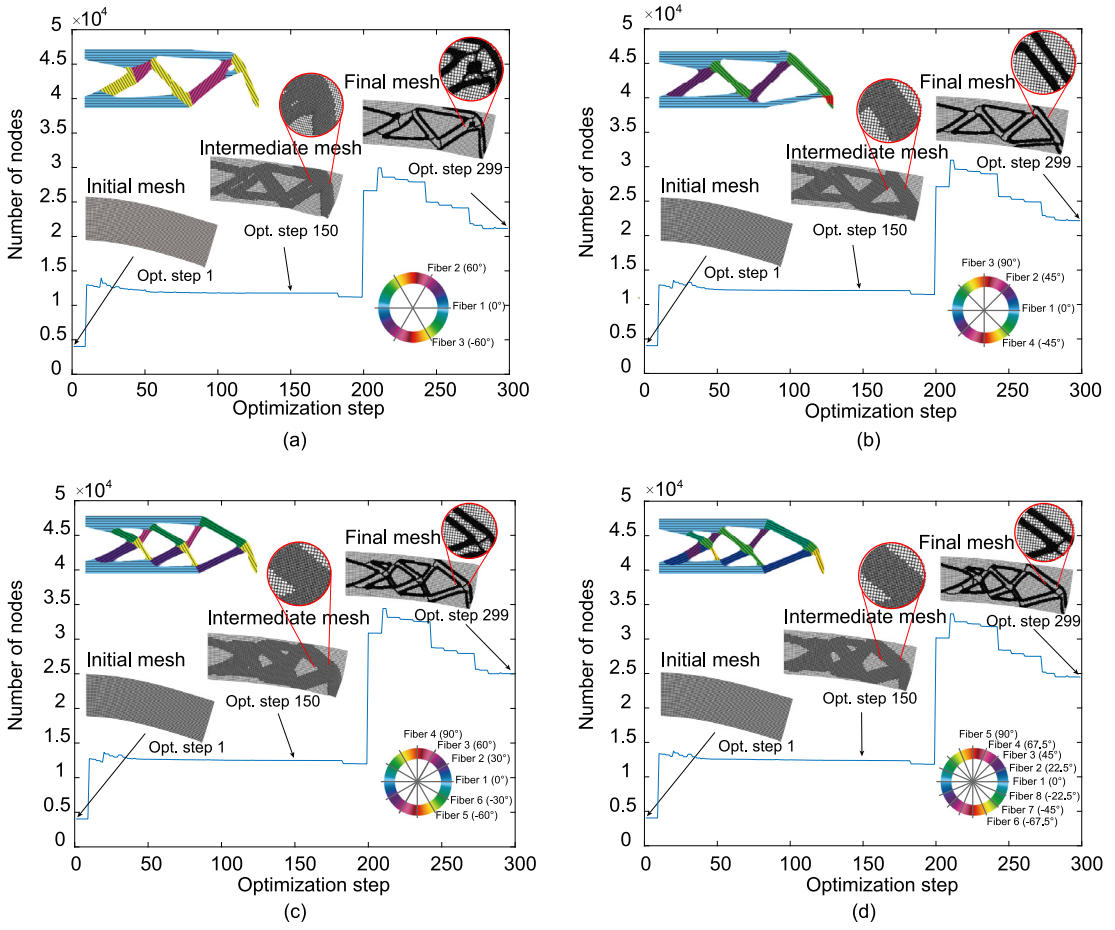


Fig. 12. Evaluations of the adaptively refined and coarsened meshes under $F = 15$ kN for (a)–(d): fiber set 1 to fiber set 4, respectively. The results presented are obtained from the BCP scheme.

whose optimized physical densities $\tilde{\rho}^{(e)}$ greater than 0.5 are counted. The histograms indicate that the maximum element-level strain in the designs optimized under $F = 50$ kN are roughly 10%, suggesting the presence of finite deformations in addition to large displacements and rotations.

The last part of this example demonstrates the computational efficiency in both state equation solver and the design variable update algorithm. We will restrict our attention to the design case of $F = 15$ kN, while the conclusions also apply to other design cases. Fig. 12(a)–(d) show the evolution of the adaptively refined and coarsened meshes under $F = 15$ kN for fiber sets 1 to 4, respectively. The results show that the adaptive refinement

Table 4
Comparison of the total wall-clock time used and the final compliance obtained by the decoupled fiber-matrix design update scheme (Eqs. (48), (50), and (51)) and the MMA (coupled) update under the load level $F = 15$ kN.

Fiber set		1		2		3		4	
Interpolation scheme		BCP	HSP	BCP	HSP	BCP	HSP	BCP	HSP
Compliance (kJ)	Decoupled fiber -matrix update	5.39	5.41	5.07	5.09	4.92	4.91	4.92	4.91
	MMA update	5.38	5.43	5.07	5.11	4.93	4.92	4.93	4.91
Wall-clock time (Sec.)	Decoupled fiber -matrix update	74.4	76.7	81.0	83.5	85.4	90.1	85.4	82.7
	MMA update	437.4	750.3	717.6	1032.6	588.4	1027.7	863.6	1403.7

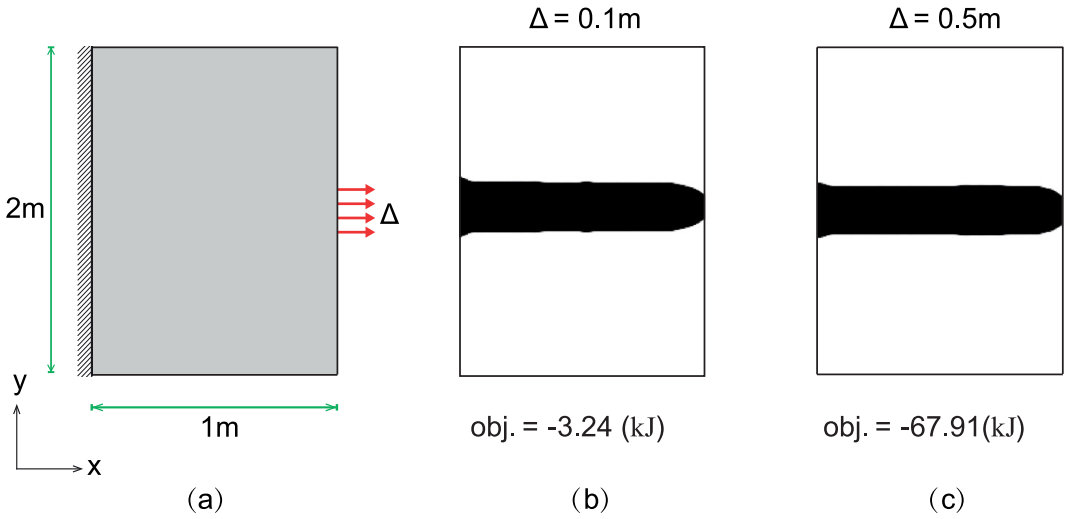


Fig. 13. (a) Design domain, load, and boundary conditions of Example 2. (b) The optimized design with its objective value obtained with $\Delta = 0.1$ m. (c) The optimized design with its objective function value obtained with $\Delta = 0.5$ m. Both designs are plotted in their undeformed configuration.

and coarsening scheme is effective in reducing the system size (as compared to uniform meshes) while maintaining high-resolution representation of the designs. Next, we compare the computational efficiency of the derived update scheme (in which the update of each design variable set is theoretically decoupled) and the standard MMA (in which the update of all the design variables is coupled). We summarize the final compliance values and the total computational time spent by the two update schemes in Table 4. The decoupled update are implemented in parallel.[1] From the comparison, we conclude that both the proposed decoupled update scheme and MMA lead to comparable compliance values, yet the proposed decoupled update scheme is more efficient and scalable with respect to the number of candidate fiber orientations.

6.2. Example 2: maximization of potential energy at the equilibrium state

The dimensions and boundary conditions are shown in Fig. 13(a). The design domain is discretized using a 50×100 initial grid, which is then adaptively refined and coarsened throughout the optimization. The adaptive refinement starts at optimization 10 and is performed every 5 optimization steps afterwards. We allow for 1

[1] The results in comparison are performed on a machine with an Intel(R) Xeon(R) Silver 4116 CPU, 2.10 GHz processor and 64 GB of RAM, running MATLAB R2018b.

Table 5
A list of parameters considered in Example 2.

Applied Disp. Δ	(i) 0.1 m, and (ii) 0.5 m
Fiber set 1	$\{30°, 45°, -30°, -45°\}$
Fiber set 2	$\{45°, 60°, -45°, -60°\}$
Volume fraction upper bound $V_{\max}$	10%
Maximum optimization step	300
Matrix Poisson's ratio ν	0.35
Fiber orientation interpolation scheme	BCP

Table 6
The moduli of matrix material and fibers considered in Example 2.

Case	Matrix E_0 (kPa)	Fiber E_t and E_c (kPa)	E_t/E_0
1	3,000	3,000	1
2	3,000	6,000	2
3	3,000	15,000	5
4	3,000	30,000	10

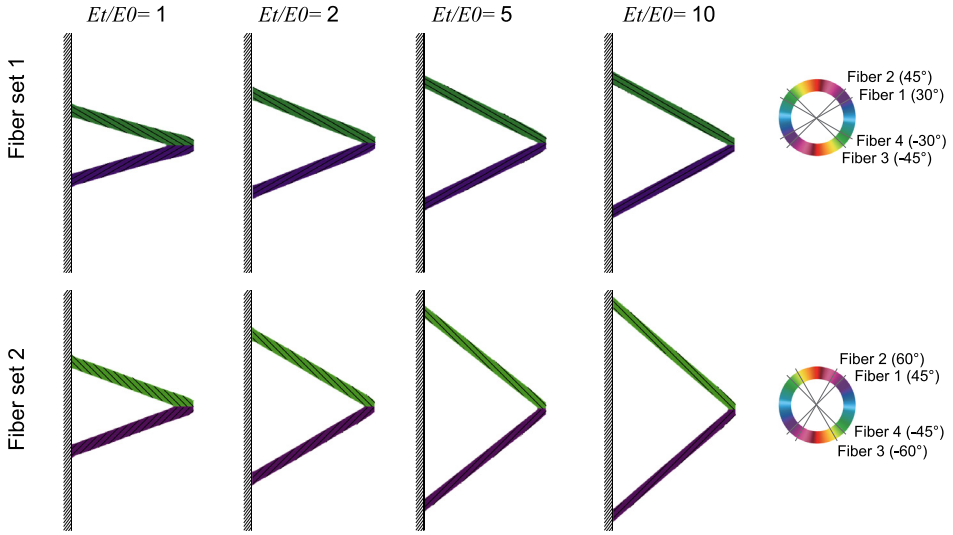


Fig. 14. Final deigns obtained for fiber sets 1 and 2 under $\Delta = 0.1$ m and with various E_t/E_0 ratios. The designs are plotted in their undeformed configurations.

refinement level before optimization step 190 and 2 refinement levels afterwards. The adaptive coarsening starts at optimization 12 and is performed every 3 optimization steps afterwards. The Newton's method with inexact line search [58,59] is adopted with a tolerance of 10^{-6}.

We adopt continuation for parameters p_ρ, p_m, β and R. The parameter p_ρ is initialized as 1 and is increased by 0.2 every step until $p_\rho = 3$ is reached. The parameter p_m is initialized as 1 and is increased by 0.1 every 2 steps until $p_m = 3$ is reached. The parameter β is initialized as 1 and, starting from optimization step 120, β is doubled every 30 steps until $\beta = 64$ is reached. The filter radius R for both density and fiber orientation variables are set to $R = 0.04$ m and, starting from optimization step 190, the filter radius for the fiber orientation variables is reduced by $\Delta R = -0.04/12$ m until $R_{\min} = 0.04/6$ m is reached. Throughout this example, we impose symmetry on the density design variables ρ_e (not the fiber orientation design variables) in the vertical direction. Other design parameters and design case setups are summarized in Tables 5 and 6, respectively.

We study two prescribed displacement Δ: (i) $\Delta = 0.1$ m and (ii) $\Delta = 0.5$ m. The optimized designs obtained by only considering the contribution from the matrix phase are shown in Fig. 13(b) and (c), respectively, for $\Delta = 0.1$ m and $\Delta = 0.5$ m. Both designs are identical in shape, and each consists of only one horizontal member. When

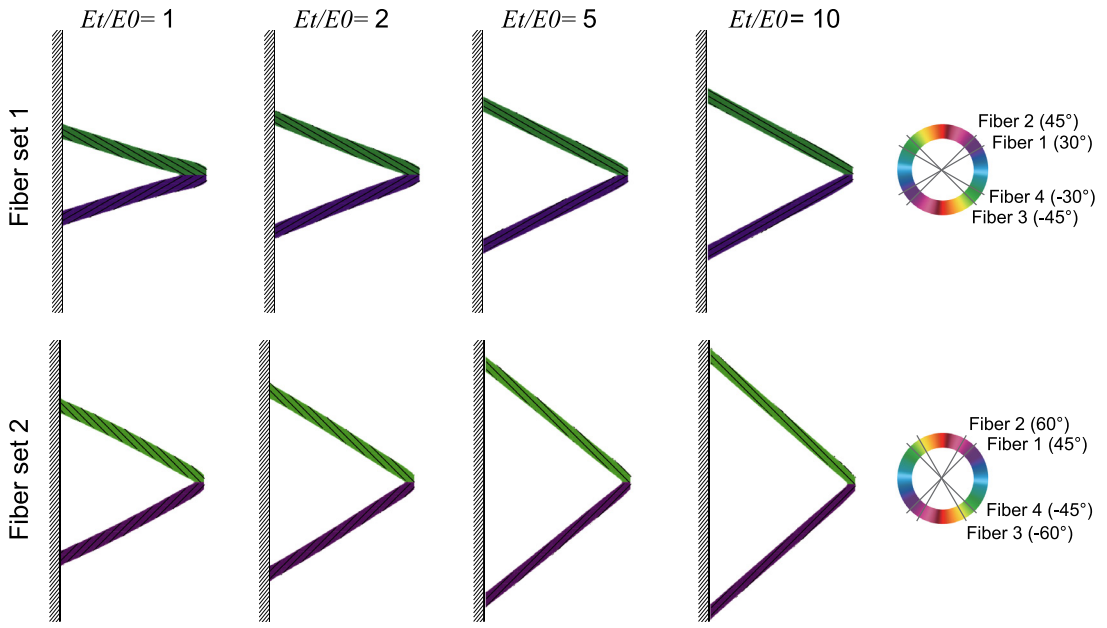


Fig. 15. Final deigns obtained for fiber sets 1 and 2 under $\Delta = 0.5$ m and with various E_t/E_0 ratios. The designs are plotted in their undeformed configurations.

we include the contribution from fiber reinforcements, we obtain designs with varying topologies depending on the relative stiffness and orientations of the fibers as well as the applied displacements Δ. Figs. 14 and 15 depict the final deigns (plotted in the undeformed configuration) obtained under $\Delta = 0.1$ m and $\Delta = 0.5$ m for fiber sets 1 and 2, respectively. In addition, a comparison of the objective function values of the optimized designs with those of the initial designs for all the design cases considered is summarized in Table 7. Similarly to the first example, the objective functions of the initial designs are evaluated on the final meshes.

From the results, several observations can be made regarding the influences of the candidate orientations, the relative stiffness of fibers to matrix, E_t/E_0, and the level of prescribed displacement Δ, on the optimized designs. First, fibers with orientations closer to the horizontal direction are always chosen in the optimized designs. For example, in the fiber set 1, orientations 30° and −30° are always chosen over the others regardless of the values of applied displacement Δ and E_t/E_0. This is because, with other design parameters fixed, a larger potential energy is achieved with fibers that are aligned closer to horizontal directions and, if the fibers are aligned horizontally, the designs have the largest potential energy. In addition, Table 7 indicates that, under the same Δ value and E_t/E_0 ratio, the designs obtained from the first fiber set always have larger potential energy values at its equilibrium state than those obtained from the second fiber set, because the candidate orientations in the former set are closer to 0° than those in the latter one.

Second, with a relatively small ratio E_t/E_0 (say $E_t/E_0 = 1$), the inclination angles of the members in the deigns are between 0° (i.e., Fig. 13) and the chosen orientations from the fiber sets (i.e., 30° and −30° for fiber set 1, and 45° and −45° for fiber set 2). This phenomenon reflects the interplay between the modulus of the isotropic matrix and that of the fibers. As we increase the ratio E_t/E_0, the stiffness of the fibers gradually becomes dominant over that of the matrix and, thus, the inclination angles of the members in the optimized designs gradually approach the orientations of the fibers. Third, with the larger prescribed displacement (i.e., $\Delta = 0.5$ m) and a relatively smaller E_t/E_0 ratio, we tend to obtain designs whose members have inclination angles closer to the chosen fiber orientations, reflecting the nonlinear behavior of the matrix phase under large deformations. For example, comparing the designs obtained under $\Delta = 0.1$ m and $\Delta = 0.5$ m for $E_t/E_0 = 1$ and fiber set 2, we can see that the members in the design obtained from $\Delta = 0.5$ m is inclined in an angle that is slightly closer to 45° than the one in the design obtained from $\Delta = 0.1$ m.

Convergence histories of the objective function values for cases of two Δ values and two fiber sets are shown in Figs. 16–17 considering $E_t/E_0 = 1$. As shown from the figures, the convergences of the objective functions

Table 7
Comparison of the objective function values (unit: kJ) of the optimized designs and those of the initial designs for all the design cases. Notice that the objective function is negative of the potential energy at equilibrium state. Therefore, a smaller final objective value corresponds to a larger potential energy at the equilibrium state.

E_t/E_0 ratio		1		2		5		10	
Fiber set		1	2	1	2	1	2	1	2
Initial designs obj.	$\Delta = 0.1$ m	−1.00	−0.91	−1.23	−1.05	−1.92	−1.39	−2.95	−1.87
	$\Delta = 0.5$ m	−25.03	−22.60	−32.86	−27.47	−55.22	−39.61	−91.88	−56.94
Optimized designs obj.	$\Delta = 0.1$ m	−4.55	−3.47	−6.12	−3.95	−10.86	−5.97	−18.67	−9.62
	$\Delta = 0.5$ m	−106.6	−78.8	−149.0	−99.3	−277.1	−161.5	−488.5	−268.2

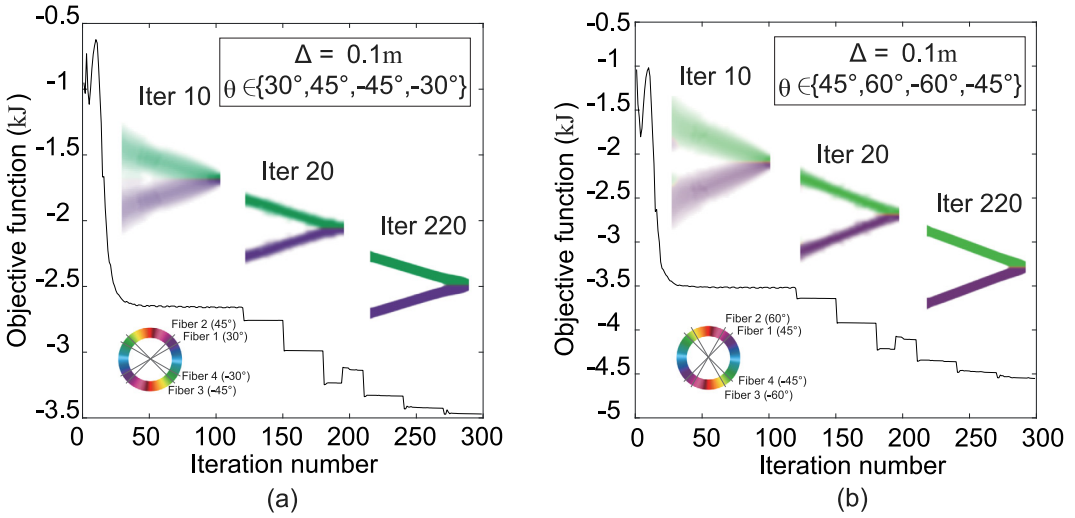


Fig. 16. Convergence history of the objective function values for $\Delta = 0.1$ m considering $E_t/E_0 = 1$, (a) fiber set 1, and (b) fiber set 2.

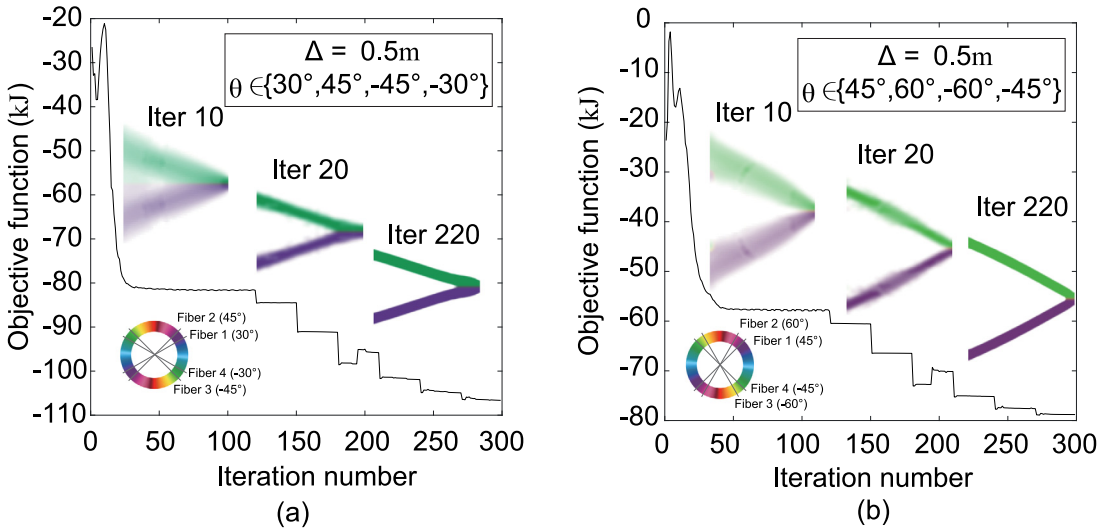


Fig. 17. Convergence history of the objective function values for $\Delta = 0.5$ m considering $E_t/E_0 = 1$, (a) fiber set 1, and (b) fiber set 2.

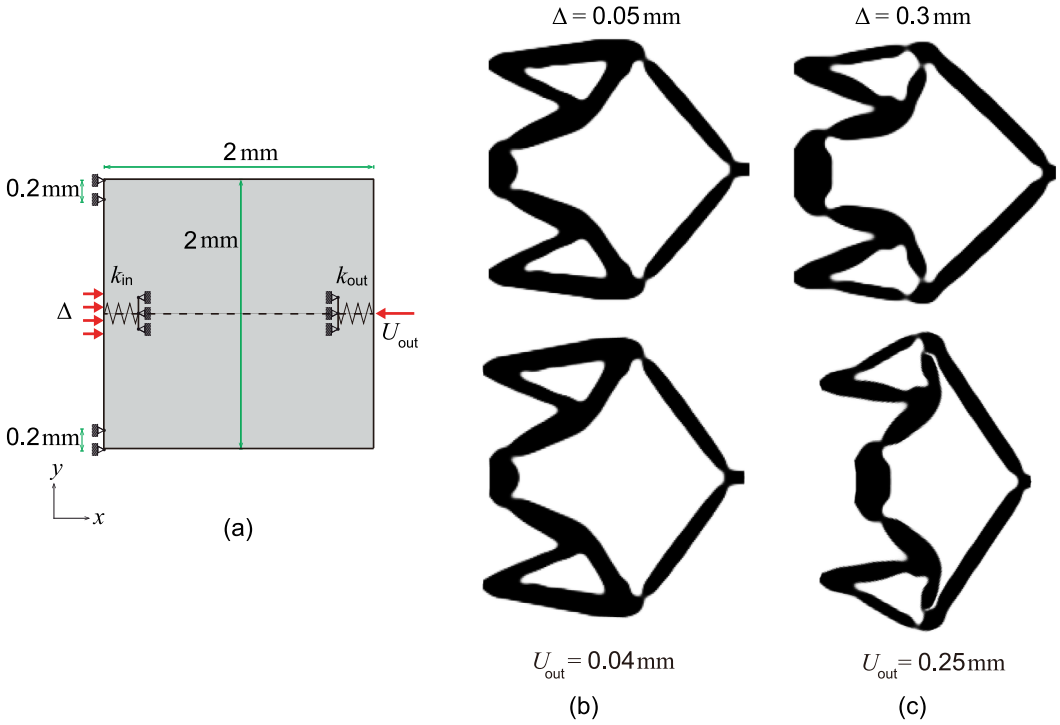


Fig. 18. (a) Design domain, load, and boundary conditions of the compliant mechanism problem. (b) The optimized design obtained under input displacement $\Delta = 0.05$ mm (Top: undeformed configuration; bottom: deformed configuration). (c) The optimized design obtained under input displacement $\Delta = 0.3$ mm (Top: undeformed configuration; bottom: deformed configuration).

are smooth except for those steps where either continuation of parameters or adaptive refinement and coarsening is performed.

6.3. Example 3: fiber-reinforced compliant mechanism

The design domain and boundary conditions are shown in Fig. 18(a). The goal is to minimize the displacement of the middle point of its right side (corresponds to maximizing the displacement of that point in the negative direction). The filter boundary padding technique [60] is adopted to address the influence of boundary effect on the optimized design due to filtering. We initially discretize the design domain with an 80×80 mesh, which is adaptively refined and coarsened throughout the optimization process. The adaptive refinement starts at optimization step 30 and is performed every 5 optimization steps. One refinement level is allowed. The adaptive coarsening starts at optimization step 30 and is performed every 3 optimization steps. The Newton's method with inexact line search [58,59] is adopted with a tolerance of 10^{-6}.

The penalization parameter p_ρ is set as 3 and we adopt continuation for parameters p_m, β, and R. The parameter p_m is initialized as 1 and increased by 0.1 starting from step 60 every 4 steps until $p_m = 3$ is reached. The parameter β is initially taken as 1 and doubled starting from step 120 every 30 steps until a maximum value of $\beta = 32$ is reached. Lastly, the initial value of the filter radius R for both the density and the fiber orientation fields is set to $R = 0.1$ mm and, starting from optimization step 190, the radius of the fiber orientation field is reduced by $\Delta R = 3R/40$ mm until a minimum value of $R_{\min} = 0.025$ mm is reached. Throughout this example, we impose symmetry on the density design variables ρ_e (but not for the fiber orientation design variables) in the vertical direction. A summary of other design parameters and design case setups considered in this example is provided in Table 8.

Fig. 18(b) and (c) show the designs (in both undeformed and deformed configurations) obtained by only considering matrix materials under $\Delta = 0.05$ mm and $\Delta = 0.3$ mm, respectively. In Figs. 19–20, we show the

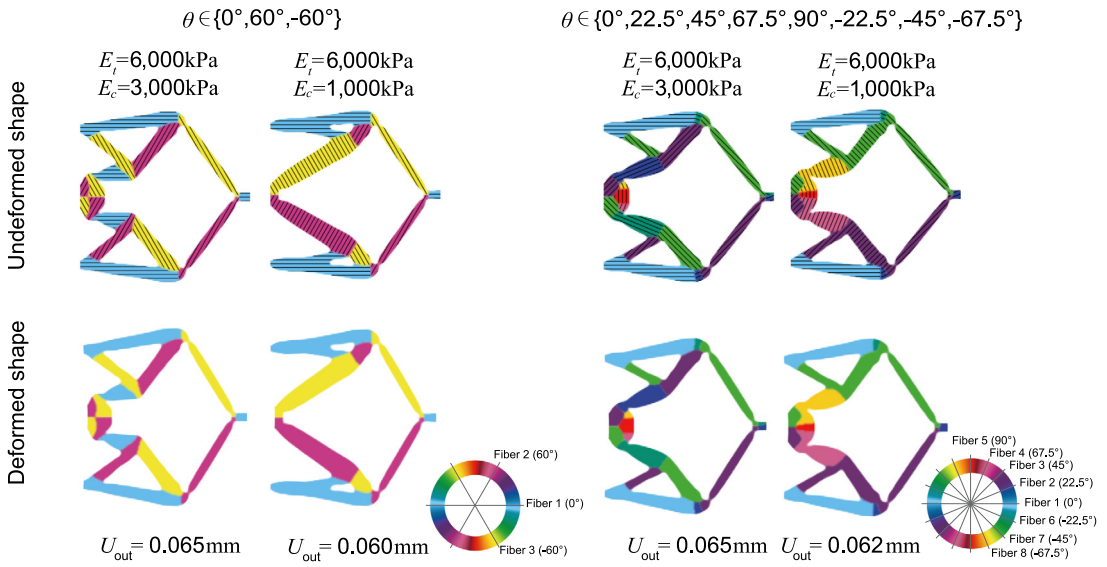


Fig. 19. Optimized compliant mechanism design under $\Delta = 0.05$ mm. Both undeformed and deformed configurations are plotted.

Table 8
A list of parameters considered in Example 3.

Applied Disp. Δ	(i) 0.05 mm, and (ii) 0.3 mm
Fiber set 1	$\{0°, 60°, -60°\}$
Fiber set 2	$\{0°, 22.5°, 45°, 67.5°, 90°, -67.5°, -45°, -22.5°\}$
Volume fraction upper bound $V_{\max}$	25%
Maximum optimization step	300
Matrix E_0 (kPa) and ν	3,000 and 0.35
Fiber E_t (kPa)	6,000
Fiber E_t/E_c ratio	Case 1: 2, Case 2: 6
Spring stiffness k_{in} and k_{out} (N/m)	200 and 60
Fiber orientation interpolation scheme	HSP

Table 9
The comparison of the output displacements, U_{out}, (unit: mm) of the optimized designs with those of the initial designs for all the design cases. Notice that U_{out} takes positive values in the negative direction of x.

E_t/E_c ratio		2		6	
Fiber set		1	2	1	2
Initial design obj.	$\Delta = 0.05$ mm	−0.003	−0.003	−0.003	−0.003
	$\Delta = 0.3$ mm	−0.012	−0.013	−0.012	−0.012
Optimized design obj.	$\Delta = 0.05$ mm	0.065	0.065	0.060	0.062
	$\Delta = 0.3$ mm	0.33	0.34	0.30	0.33

optimized designs (in both undeformed and deformed configurations) obtained under $\Delta = 0.05$ mm and $\Delta = 0.3$ mm, respectively, for all the design cases considered. Additionally, a comparison of the objective function values of the optimized designs with those of the initial designs for all the design cases is summarized in Table 9. Similarly to the previous examples, the objective functions of the initial designs are evaluated on the final meshes.

Several observations can be made. First, for all the cases considered in this example, incorporating fiber reinforcements in compliant mechanism designs yields designs with larger output displacement compared to those with only matrix materials. Second, the candidate orientations in the fiber set has an apparent influence on the topology optimization design. For both sets of fibers with different E_t/E_c ratios and both input displacements Δ, the member orientations in the designs obtained by considering fiber set 1 (with 3 candidate orientations) are different

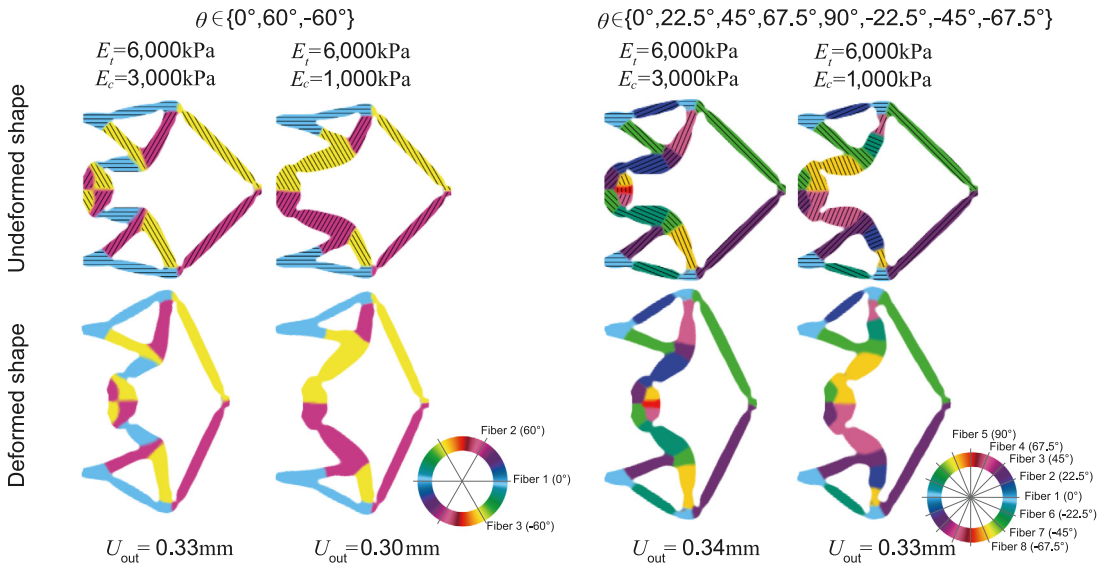


Fig. 20. Optimized compliant mechanism design under $\Delta = 0.3$ mm. Both undeformed and deformed configurations are plotted.

from those obtained by using fiber set 2 (with 8 candidate orientations) in several regions of the designs. In addition, the optimized designs obtained by considering fiber set 2 resemble more to the ones obtained by considering the matrix phase only. This indicates that, by increasing the number of candidate orientations of fibers in the proposed formulation, we could obtain more similar designs (in terms of topology) to the ones by using isotropic matrix. Third, the nonlinear behavior of fibers in tension and compression considerably affects the distribution of fiber orientations in the optimized designs. For both Δ values, the fibers in the designs obtained assuming with $E_t/E_c = 2$ are aligned closely with the longitudinal directions of their underlying structural members. However, when the ratio E_t/E_c is increased to 6, the fibers tend to align differently in structural members under tension and compression. When a structural member is in tension, its fiber reinforcements are oriented closely to its longitudinal direction. On the contrary, when a structural member is in compression, its fiber reinforcements are oriented closely to its transverse direction. Because the stiffness of fibers is much higher in tension than in compression when $E_t/E_c = 6$, such arrangement of fibers ensures that they are always used to resist tensile deformations (namely, the longitudinal deformation of the structural members in tension and the transverse deformation of those in compression).

Next, Fig. 21 plots the input displacement Δ vs. output displacement U_{out} for the optimized fiber-reinforced compliant mechanism designs. We can see that, as compared to the ones of the optimized designs obtained by $\Delta = 0.3$ mm, the input displacement vs. output displacement curves of the optimized designs obtained by $\Delta = 0.05$ mm exhibit softer (i.e., smaller U_{out}) behaviors when subjected to a large input displacement values such as $\Delta = 0.25$ mm, showing the influences of nonlinearity and large deformations. In addition, Fig. 22(a)–(d) plots histograms of the principal stretches of each element in the designs optimized under $\Delta = 0.3$ mm for various design cases. Only those elements whose optimized physical densities $\tilde{\rho}^{(e)}$ greater than 0.5 are counted. Similar to the ones presented in Example 1, those histograms indicate that the maximum element-level strain in the optimized designs obtained under $\Delta = 0.3$ mm are roughly 10%, suggesting the presence of finite deformations in addition to large displacements and rotations.

Finally, we remark that some of our optimized designs show hinge connections, which could be alleviated using the robust formulation [61]. This will be subjected to future research.

7. Concluding remarks

We present a general topology optimization framework for the computational optimized design of hyperelastic structures reinforced by nonlinear and anisotropic fibers under large deformations. We introduce a design parameterization in which the matrix phase and fiber orientations are represented by two separate sets of design variables,

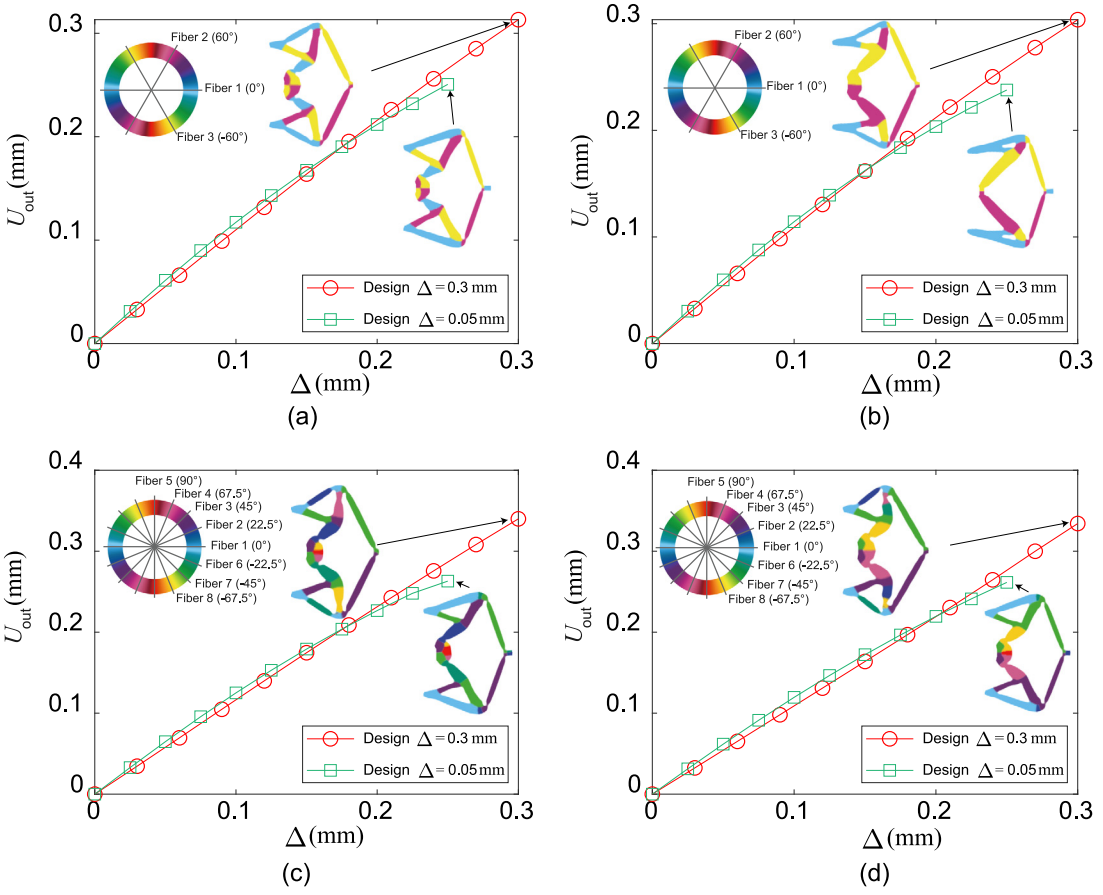


Fig. 21. The input displacement Δ vs. output displacement U_{out} curves of the fiber-reinforced optimized compliant mechanism designs obtained using (a) fiber set 1 and $E_t/E_c = 2$, (b) fiber set 2 and $E_t/E_c = 2$, (c) fiber set 1 and $E_t/E_c = 6$, and (d) fiber set 2 and $E_t/E_c = 6$. The optimized designs obtained under both $\Delta = 0.05$ mm and $\Delta = 0.3$ mm are loaded until either $\Delta = 0.3$ mm or the solver convergence failure is reached.

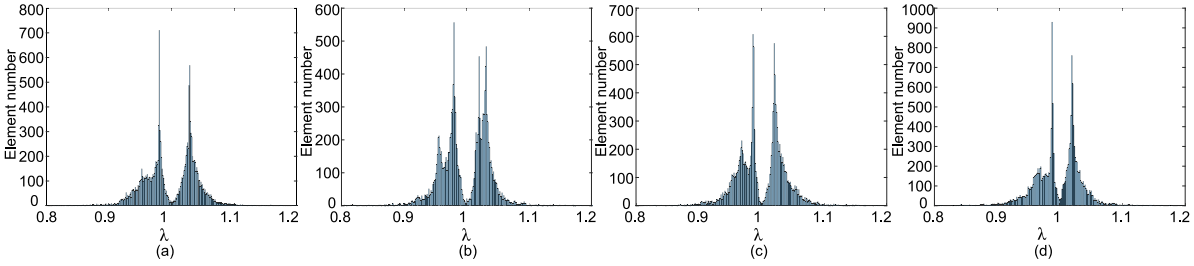


Fig. 22. The principal stretch levels within each elements are quantified for the optimized designs obtained by $\Delta = 0.3$ mm: (a) fiber set 1 and $E_t/E_c = 2$, (b) fiber set 2 and $E_t/E_c = 2$, (c) fiber set 1 and $E_t/E_c = 6$, and (d) fiber set 2 and $E_t/E_c = 6$. Only those elements whose optimized physical densities $\tilde{\rho}^{(e)}$ greater than 0.5 are counted.

thus, it allows for the direct incorporation of geometric (e.g., symmetry) constraints on the matrix phase without influencing the orientation distribution of the underlying fiber phase. An anisotropic material interpolation scheme is proposed to integrate contributions from both the matrix and the fiber phases into the stored-energy function of the resulting transversely isotropic system. We demonstrate that the design parameterization and the anisotropic material interpolation scheme are compatible with various fiber field interpolation schemes (e.g., BCP and HSP)

and effectively account for up to 8 candidate fiber orientations. An energy interpolation scheme is incorporated to overcome the convergence difficulty of the low-density elements during the optimization and is shown to be robust under large deformations through numerical examples. This work introduces several strategies to improve computational efficiency in both optimization and simulations aspects without sacrificing design accuracy. On the optimization side, we derive a decoupled design variable update scheme, in which the update of design variables associated with the matrix field and the update of each fiber orientation field are theoretically decoupled and performed in parallel. We further derive explicit formula (i.e., no iteration needed) for the update of the design variables associated with the fiber orientation fields. We demonstrate the computational efficiency of the proposed decoupled fiber-matrix update scheme over a commonly-used scheme (MMA) in Table Table 4. On the simulation side, the VEM with adaptive mesh refinement and coarsening are employed to efficiently solve the nonlinear state equations. The use of the VEM to solve transversely isotropic finite elasticity boundary value problems is verified through a convergence study, which demonstrates the convergence of numerical solutions under mesh refinement.

We present three design examples to demonstrate that the proposed topology optimization framework effectively accounts for various design objectives and captures the influences of the candidate fiber orientations and nonlinear fiber behaviors on the topology of the optimized designs. The main conclusions are summarized as follows:

- Accounting for fiber reinforcements for the matrix materials leads to designs with improved performance as compared to the designs consist of only matrix materials. Although problem-dependent exceptions could occur due to the non-convex nature of the formulation, it appears to be a general trend that considering more number of candidate fiber orientations in the proposed formulation leads to designs with better objective function values as results of enriched design spaces.
- The number and distribution of candidate fiber orientations have apparent influences on the topologies of the optimized designs. With few candidate fiber orientations (e.g., 3 or 4), we obtain designs that are different from the ones obtained by considering only (isotropic) matrix phase. With more number of candidate fiber orientations (i.e., 8), the topologies of the optimized designs resemble the ones obtained by considering only (isotropic) matrix phase.
- The relative stiffness of the fibers to the matrix phase also influences the topologies of the optimized designs. When the stiffness of the fibers is high compared to that of the matrix (e.g., $E_t/E_0 = 10$), the fibers and their underlying structural members in the optimized designs typically align in the same orientations. When the stiffness of the fibers is comparable to that of the matrix (e.g., $E_t/E_0 = 1$), the fibers and their underlying structural members may be aligned in dissimilar orientations in the optimized design.
- The nonlinear behavior of fibers in tension and compression does not appear to have significant influence on the orientations of their underlying structural members but can greatly influence their distributions in the optimized design. When fibers with relatively large E_t/E_c ratios (e.g. $E_t/E_c = 10$) are considered, they tend to align longitudinally when the underlying structural members are under tension and transversely when the underlying structural members are under compression. When fibers with relatively small E_t/E_c ratios (e.g., $E_t/E_c = 2$) are considered, they always align in (or close to) longitudinal directions of their underlying structural members.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Sensitivity analysis

This appendix presents the sensitivity analysis of the optimization problem (22). We first perform the sensitivity analysis of the objective function J. In the following discussions, we keep J in its general form. The sensitivity of J with respect to the filtered design variable $\bar{\rho}^{(e)}$ can be computed via the chain rule and adjoint method [56] as follows,

$$\frac{\partial J}{\partial \bar{\rho}^{(e)}} = \left\{ \frac{\partial J(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \mathbf{u}_h)}{\partial \tilde{\rho}^{(e)}} + \boldsymbol{\lambda}^T \left[\frac{\partial \mathbf{R}\left(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \boldsymbol{\gamma}, \mathbf{u}_h\right)}{\partial w^{(e),\rho}} + \frac{\partial \mathbf{R}\left(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \boldsymbol{\gamma}, \mathbf{u}_h\right)}{\partial \gamma^{(e)}} \frac{\partial \gamma^{(e)}}{\partial w^{(e),\rho}} \right] \frac{\partial w^{(e),\rho}}{\partial \tilde{\rho}^{(e)}} \right\} \frac{\partial \tilde{\rho}^{(e)}}{\partial \bar{\rho}^{(e)}}, \tag{55}$$

where $\partial \gamma^{(e)}/\partial w^{(e),\rho}$, $\partial w^{(e),\rho}/\partial \tilde{\rho}^{(e)}$ and $\partial \tilde{\rho}^{(e)}/\partial \bar{\rho}^{(e)}$ are given by

$$\frac{\partial \gamma^{(e)}}{\partial w^{(e),\rho}} = \frac{\beta_\gamma \left[\tanh\left(\beta_\gamma \left(w^{(e),\rho} - \rho_\gamma\right)\right)^2 - 1 \right]}{\tanh\left(\beta_\gamma(\rho_\gamma - 1)\right) - \tanh\left(\beta_\gamma \rho_\gamma\right)} \tag{56}$$

$$\frac{\partial w^{(e),\rho}}{\partial \tilde{\rho}^{(e)}} = p_\rho (1-\varepsilon) \left(\tilde{\rho}^{(e)}\right)^{p_\rho - 1} \tag{57}$$

$$\frac{\partial \tilde{\rho}^{(e)}}{\partial \bar{\rho}^{(e)}} = \frac{\beta \left[\tanh\left(\beta \bar{\rho}^{(e)} - \rho_0\right)^2 - 1 \right]}{\tanh\left(\beta(\rho_0 - 1)\right) - \tanh\left(\beta \rho_0\right)}, \tag{58}$$

and $\boldsymbol{\lambda}$ is the adjoint vector which can be obtained by solving the following adjoint system,

$$\boldsymbol{\lambda} = -\mathbf{K}_T^{-1} \left(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \boldsymbol{\gamma}, \mathbf{u}_h\right) \frac{\partial J(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \mathbf{u}_h)}{\partial \mathbf{u}_h}, \tag{59}$$

where $\mathbf{K}_T \doteq \partial \mathbf{R}/\partial \mathbf{u}_h$ is the global tangent stiffness matrix evaluated at the equilibrating displacement field $\mathbf{u}_h$.

Similarly, the derivatives of J with respect to the filtered design variable $\bar{\xi}_i^{(e)}$, $i = 1, \ldots, m_\xi$, are given by

$$\frac{\partial J}{\partial \bar{\xi}_i^{(e)}} = \frac{\partial J(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \mathbf{u}_h)}{\partial \bar{\xi}_i^{(e)}} + \boldsymbol{\lambda}^T \sum_{j=1}^{m} \frac{\partial \mathbf{R}\left(\boldsymbol{\rho}, \boldsymbol{\xi}_1, \ldots, \boldsymbol{\xi}_{m_\xi}, \boldsymbol{\gamma}, \mathbf{u}_h\right)}{\partial w_j^{(e),m}} \frac{\partial w_j^{(e),m}}{\partial m_j^{(e)}} \frac{\partial m_j^{(e)}}{\partial \bar{\xi}_i^{(e)}}, \tag{60}$$

where $\boldsymbol{\lambda}$ is the adjoint variable solved by Eq. (59), $\partial w_j^{(e),m}/\partial m_j^{(e)}$ are given by

$$\frac{\partial w_j^{(e),m}}{\partial m_j^{(e)}} = p_m \left(m_j^{(e)}\right)^{(p_m - 1)}, \tag{61}$$

and the detailed expressions of $\partial m_j^{(e)}/\partial \bar{\xi}_i^{(e)}$ are given by Eqs. (28) and (31) for BCP and HSP schemes, respectively. For brevity, the explicit expressions of $\partial \mathbf{R}/\partial w^{(e),\rho}$, $\partial \mathbf{R}/\partial w_j^{(e),m}$ are not provided here.

Once $\partial J/\partial \bar{\rho}^{(e)}$, $\partial J/\partial \bar{\xi}_i^{(e)}$, $i = 1, \ldots, m_\xi$, are obtained, the derivatives of J with respect to the unfiltered design variables are given accordingly by

$$\frac{\partial J}{\partial \rho^{(e)}} = \sum_{\ell=1}^{M} [\mathbf{H}]_{\ell e} \frac{\partial J}{\partial \bar{\rho}^{(\ell)}} \quad \text{and} \quad \frac{\partial J}{\partial \xi_i^{(e)}} = \sum_{\ell=1}^{M} [\mathbf{H}]_{\ell e} \frac{\partial J}{\partial \bar{\xi}_i^{(\ell)}}, \quad i = 1, \ldots, m_\xi, \tag{62}$$

where $[\mathbf{H}]_{\ell e}$ stands for the (ℓ, e)th component of the filter matrix $\mathbf{H}$. For the sensitivities of the volume constraint g_V, we obtain the following expression,

$$\frac{\partial g_V}{\partial \rho^{(e)}} = \sum_{\ell=1}^{M} [\mathbf{H}]_{\ell e} \left[V^{(\ell)} \frac{\partial \tilde{\rho}^{(\ell)}}{\partial \bar{\rho}^{(\ell)}} \right], \tag{63}$$

where $\partial \tilde{\rho}^{(\ell)}/\partial \bar{\rho}^{(\ell)}$ is given by (58).

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